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PaLI-X: On Scaling up a Multilingual Vision and Language Model

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Abstract

We present the training recipe and results of scaling up PaLI-X, a multilingual vision and language model, both in terms of size of the components and the breadth of its training task mixture. Our model achieves new levels of performance on a wide-range of varied and complex tasks, including multiple image-based captioning and question-answering tasks, image-based document understanding and few-shot (in-context) learning, as well as object detection, video question answering, and video captioning. PaLI-X advances the state-of-the-art on most vision-and-language benchmarks considered (25+ of them). Finally, we observe emerging capabilities, such as complex counting and multilingual object detection, tasks that are not explicitly in the training mix.

1 Introduction

The success of scaling language models [1, 2, 3, 4] makes it appealing to similarly scale Vision-Language (V&L) models, and investigate the improvements, capabilities, and emergent properties of such models. Inspired by the work in [5], we present PaLI-X, a multilingual vision and language model with reusable scaled-up components, consisting of a pretrained large-capacity visual encoder (using [6] as the starting point) and a pretrained language-only encoder-decoder (using [7] as the starting point), further trained at-scale on a vision-and-language data mixture using a combination of self-supervision and full-supervision signals.

One clear pattern that emerges from the combination of results from PaLI [5] and the work we present in this paper is that scaling *both* V&L components together brings increases in performance across a wide range of tasks. We show this by comparing against the same benchmarks used for PaLI (Fig. 1, Left), and also against new benchmarks for which the new capabilities of PaLI-X are evaluated (e.g., ChartQA, AI2D, DocVQA, InfographicVQA, as well as video understanding tasks). We observe that scaling leads to large improvements over the results of the PaLI model, and also over specialized large-scale models that are trained specifically to solve certain tasks, often with the help of (often much larger) text-only LLMs [8]. In particular, we find that increasing both the effective capacity of the vision component (which [9] does more unilaterally) and of the language component

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PaLI-X：扩展多语言视觉与语言模型

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摘要

我们介绍了 PaLI-X 的训练方案与结果，该模型是一个多语言视觉与语言模型，在组件规模以及训练任务组合的广度上都进行了扩展。我们的模型在各种复杂任务上达到了新的性能水平，包括多种基于图像的图像描述和问答任务、基于图像的文档理解和少样本（上下文内）学习，以及目标检测、视频问答和视频描述。PaLI-X 在所考察的大多数视觉与语言基准（其中 25+ 个）上推进了最先进水平。最后，我们观察到了涌现能力，例如复杂计数和多语言目标检测，这些任务并未明确包含在训练组合中。

1 引言

语言模型扩展的成功 [1, 2, 3, 4] 使得以类似方式扩展视觉-语言（V&L）模型颇具吸引力，并可研究此类模型的改进、能力和涌现特性。受 [5] 中工作的启发，我们提出了 PaLI-X，这是一个多语言视觉与语言模型，具有可复用的扩展组件，由一个预训练的大容量视觉编码器（以 [6] 作为起点）和一个预训练的纯语言编码器-解码器（以 [7] 作为起点）组成，并进一步在视觉与语言数据混合集上，结合自监督和全监督信号进行大规模训练。

从 PaLI [5] 的结果与本文所呈现工作的结合中浮现出的一个清晰模式是，同时扩展 *both* V&L 组件会在广泛任务上带来性能提升。我们通过与 PaLI 所使用的相同基准进行比较（图 1，左）来展示这一点，也通过与用于评估 PaLI-X 新能力的新基准进行比较来展示这一点（例如 ChartQA、AI2D、DocVQA、InfographicVQA，以及视频理解任务）。我们观察到，扩展不仅相较于 PaLI 模型的结果带来了大幅改进，也优于那些专门为解决特定任务而训练的专用大规模模型；这些模型通常借助（往往大得多的）纯文本 LLM [8]。特别是，我们发现，同时提高视觉组件（[9] 更偏向单方面这样做）的有效容量和语言组件的有效容量

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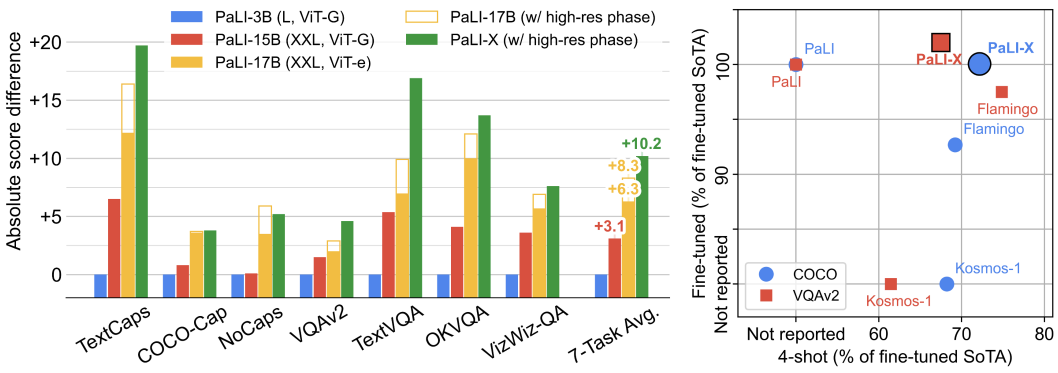


Figure 1: [Left] Comparing PaLI-X against PaLI on image-captioning and VQA benchmarks. [Right] The Pareto frontier between few-shot and fine-tuned performance, comparing PaLI-X with PaLI [5], Flamingo [10], and Kosmos-1 [11].

(which [10] also does unilaterally) is beneficial; the new PaLI-X model provides more balanced parameter allocation than any other prior work (roughly 40%-60% split of the total capacity).

Aside from confirming the impact of scale, the original contribution of PaLI-X consists in leveraging the mixture-of-objectives proposed in [7] for vision-and-language modeling, and showing that it results in a model that improves both state-of-the-art results and the Pareto frontier for fine-tuning and few-shot configurations (Fig. 1, Right).

We also observe emergent properties based on PaLI-X's results compared to previous models with similar architecture but smaller sizes. For instance, we report drastically improved performance on the counting ability (See Table 1 and Appendix B), both for the plain variety (count all instances of a class) and the complex variety (count instances based on a natural language description), that are not attributable to training design[1]. Additionally, we present qualitative insights into the model's performance (Appendix A), with an emphasis on multilingual transfer learning such as the ability to detect objects using non-English labels (Fig. 2), and the ability to switch between the language of text present in the image (e.g., English) and the language of the generated image caption (e.g., Romanian).

Our technical contributions include the following:

1. We scale a Vision-Language model to achieve outstanding performance on a wide variety of benchmarks. We observe that scaling *both* the Vision & Language components is advantageous and report that performance remains unsaturated at this scale.
2. We show that training such a model with a mixture of objectives that combines prefix-completion and masked-token completion improves the Pareto frontier for fine-tuning vs few-shot performance at this scale.
3. We show that a high-capacity vision encoder (ViT-22B) can be effectively co-trained for image classification and OCR label classification[2] to achieve significant improvements on V&L tasks for which the understanding of text-within-image is crucial.
4. Overall, PaLI-X improves SoTA results via fine-tuning on 15+ benchmarks, and we show that it is the first of its kind to simultaneously adapt via multitask fine-tuning to a diverse set of benchmarks without significant performance degradation.

2 Related Work

Similar to large language models such as GPT4 [12] and PaLM [1], the benefit of scale has also been observed in recent vision and vision-language models. Flamingo [10] used a frozen language

[1]Plain counting is usually achievable via good object detection, while complex counting requires a fine-grained understanding of the alignment between language-based specifications and visually-based occurrences.

[2]We use OCR tokens produced by the GCP Vision API over the training images as targets.

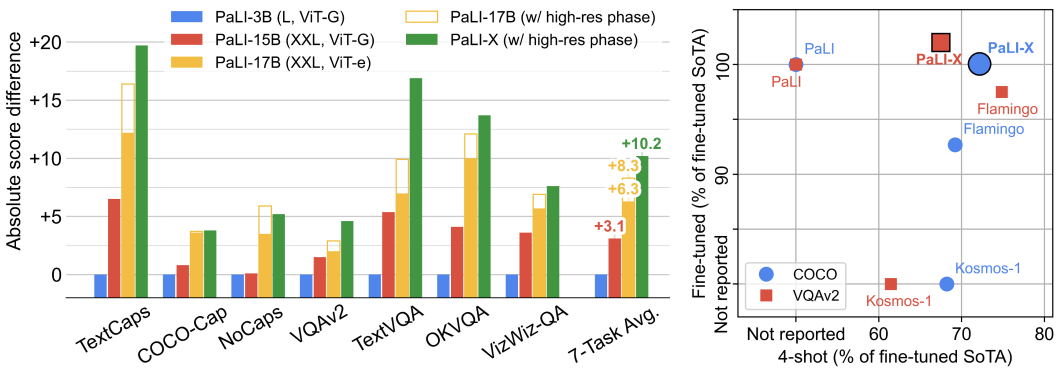


图 1：[左] 在图像字幕生成和 VQA 基准上比较 PaLI-X 与 PaLI。[右] 少样本性能与微调性能之间的帕累托前沿，比较 PaLI-X 与 PaLI [5]、Flamingo [10] 和 Kosmos-1 [11]。

（[10] 也以单方面方式这样做）是有益的；新的 PaLI-X 模型相比任何以往工作都提供了更均衡的参数分配（总容量大约按 40%-60% 划分）。

除了确认规模的影响之外，PaLI-X 的原创贡献在于利用 [7] 中提出的目标混合方法进行视觉-语言建模，并表明由此得到的模型在微调和少样本配置下均提升了当前最先进结果和帕累托前沿（图 1，右）。

我们还观察到，基于 PaLI-X 的结果，与架构相似但规模更小的先前模型相比，出现了一些涌现特性。例如，我们报告了其在计数能力上的显著性能提升（见表 1 和附录 B），包括简单类型（计算某一类别的所有实例）和复杂类型（根据自然语言描述计算实例），这些提升不能归因于训练设计[1]。此外，我们还对模型性能提供了定性见解（附录 A），重点关注多语言迁移学习，例如使用非英语标签检测物体的能力（图 2），以及在图像中出现的文本语言（例如英语）与生成的图像标题语言（例如罗马尼亚语）之间进行切换的能力。

我们的技术贡献包括以下内容：

1. 我们扩展了一个视觉-语言模型，使其在多种基准测试上实现了卓越性能。我们观察到，同时扩展*both*视觉与语言组件是有益的，并报告在这一规模下性能仍未饱和。2. 我们表明，在这一规模下，使用结合前缀补全和掩码词元补全的混合目标来训练此类模型，可以改善微调性能与少样本性能之间的帕累托前沿。3. 我们表明，高容量视觉编码器（ViT-22B）可以有效地联合训练图像分类和 OCR 标签分类[2]，从而在那些理解图像中文字至关重要的 V&L 任务上取得显著提升。4. 总体而言，PaLI-X 通过在 15+ 个基准测试上进行微调提升了 SoTA 结果，并且我们表明，它是首个能够通过多任务微调同时适应一组多样化基准测试且不出现显著性能下降的同类模型。

2 相关工作

与 GPT4 [12] 和 PaLM [1] 等大型语言模型类似，规模带来的益处也在近期的视觉和视觉-语言模型中得到了观察。Flamingo [10] 使用了一个冻结的语言

[1]Plain counting is usually achievable via good object detection, while complex counting requires a fine-grained understanding of the alignment between language-based specifications and visually-based occurrences.

[2]We use OCR tokens produced by the GCP Vision API over the training images as targets.

component and demonstrated the benefit of scaling up this part up to 70B parameters on the few-shot multimodal capabilities, while the vision encoder is fixed with 435M parameters. GIT [9], on the other hand, explored scaling of the vision component up to 4.8B parameter, with a 300M parameter language decoder. PaLI [5] explored jointly scaling the vision and language component, to 4B and 17B, respectively, and showed that scaling both components benefits a wide range of vision-language tasks. All these models took advantage of vision and language unimodal pretrained models as backbones to start multimodal training. Recently, on the vision model side, a vision transformer with 22B parameter has been introduced [6]. In this work, we make use of a ViT-22B model specifically tuned for OCR capability to explore scaling Vision-Language models to even larger parameter regime.

As first shown in [13], *large* language models are sometimes able to solve new unseen tasks at inference as long as a few examples –or *shots*– are provided as inputs. This is usually referred to as in-context learning [14]. Follow-up work proposed improved ways to split and prompt the shots, such as Chain of Thought [15] or Least-to-Most prompting [16]. So far, the vast majority of this work has been done in the context of language inputs [17]. In this work, we explore multimodal in-context learning with pairs of images and captions. Our work is aligned in spirit to Flamingo [10] that uses interleaved image text pairs in the same web page and in-context tuning [18] during pre-training. We first group the image-text pairs by url and split each group to a "shots" set and a "target" set. Then we use the few examples in the "shots" set as input features to predict the examples in the target set.

Besides solving vision-language tasks in multiple domains, recent VLMs also attempted solving these tasks at once instead of fine-tuning on each individual benchmark. Unified-IO [19] performed multitask fine-tuning and reported solid results across 16 benchmarks. Spotlight [20] reported that inside the UI domain, multitask fine-tuning can achieve a performance close to task-specific fine-tuning. In this work, we show that PaLI-X can be simultaneously fine-tuned with a diverse set of benchmarks in multiple domains without performance degradation.

3 Model

3.1 Architecture

The PaLI-X model architecture follows the encoder-decoder architecture: image(s) are processed by a ViT encoder, with the resulting visual embeddings fed to an encoder-decoder backbone, along with embeddings from additional text input (e.g., question / prefix / prompt). More details are provided in Appendix A.

Visual component Our visual backbone is scaled to 22B parameters, as introduced by [6], the largest dense ViT model to date. To equip the model with a variety of complex vision-language tasks, we specifically focus on its OCR capabilities. To that end, we incorporate an OCR-based pretraining as follows: images from the WebLI dataset [5] are annotated with OCR-text detected by GCP Vision API; the encoder is then further pre-trained with a mixture of the original JFT-based classification task and a new OCR-based classification task (whether or not a given token occurred in the image according to OCR results). See Appendix A for additional details on the visual component. PaLI-X is designed to take $n >= 1$ images as inputs (for few-shot and video understanding), with tasks involving a single image as the $n = 1$ case. For $n > 1$, each image is independently processed by the ViT module, and the patch-level embeddings coming out of ViT are flattened and concatenated to form the visual input (See Appendix A). Note that similar to the single-image case, there is no pooling over the spatial dimension before visual embeddings are aggregated over the temporal dimension. That is, for an n-frame input with k-patches per frame, the resulting visual input has $n * k$ tokens.

Overall model The encoder-decoder backbone is initialized from a variant of the UL2 [7] encoder-decoder model that uses 32B parameters. The architecture of this variant has 50 layers in both encoder and decoder (up from 32 layers in [7]), and is pretrained on a mixture of text data similar to [7]. The visual embeddings, after going through a projection layer, are concatenated with the token embeddings of the text input, and fed to the encoder-decoder backbone. Most of the pretraining tasks (with the exception of the masked image token task) predict text-only output from this multimodal input. The text input to the model typically consists of a prompt that marks what type of task it is (e.g., "*Generate caption in* $\langle \text{lang} \rangle$" for captioning tasks) and encode necessary textual input for the task (e.g., "*Answer in* $\langle \text{lang} \rangle$*: {question}*" for VQA tasks). For tasks that need OCR capabilities, we experiment with either relying solely on the text-encoding capabilities of the vision encoder, or optionally including tokens extracted by an upstream OCR system fed as additional text inputs.

组件，并展示了将这一部分扩展到 700 亿参数对少样本多模态能力的益处，而视觉编码器则固定为 4.35 亿参数。另一方面，GIT [9] 探索了将视觉组件扩展到 48 亿参数，并使用 3 亿参数的语言解码器。PaLI [5] 探索了对视觉和语言组件进行联合扩展，分别达到 40 亿和 170 亿参数，并表明同时扩展这两个组件有益于广泛的视觉-语言任务。所有这些模型都利用了视觉和语言单模态预训练模型作为骨干网络来开始多模态训练。最近，在视觉模型方面，已推出了一个拥有 220 亿参数的视觉 Transformer [6]。在这项工作中，我们使用了一个专门针对 OCR 能力进行调优的 ViT-22B 模型，以探索将视觉-语言模型扩展到更大的参数规模。

正如 [13] 首次所示，只要提供少量示例——或 *shots*——作为输入，*large* 语言模型有时能够在推理时解决新的未见任务。这通常被称为上下文学习 [14]。后续工作提出了改进的样本拆分和提示方式，例如思维链 [15] 或 Least-to-Most 提示 [16]。到目前为止，这方面的绝大多数工作都是在语言输入的背景下完成的 [17]。在这项工作中，我们探索了基于图像和标题配对的多模态上下文学习。我们的工作在精神上与 Flamingo [10] 一致，后者在预训练期间使用同一网页中的交错图像-文本对以及上下文调优 [18]。我们首先按 url 对图像-文本对进行分组，并将每组拆分为“shots”集合和“target”集合。然后，我们使用“shots”集合中的少量示例作为输入特征，来预测目标集合中的示例。

除了在多个领域解决视觉-语言任务外，近期的 VLM 还尝试一次性解决这些任务，而不是在每个单独的基准上进行微调。Unified-IO [19] 进行了多任务微调，并报告了在 16 个基准上的稳健结果。Spotlight [20] 报告称，在 UI 领域内，多任务微调可以达到接近任务特定微调的性能。在这项工作中，我们表明 PaLI-X 可以同时在多个领域的一组多样化基准上进行微调，而不会导致性能下降。

3 模型

3.1 架构

PaLI-X 模型架构遵循编码器-解码器架构：图像由 ViT 编码器处理，生成的视觉嵌入会与额外文本输入（例如问题/前缀/提示）的嵌入一起输入到编码器-解码器主干网络中。更多细节见附录 A。

视觉组件 我们的视觉骨干网络扩展到 22B 参数，如 [6] 所介绍，这是迄今为止最大的密集型 ViT 模型。为了使模型能够处理各种复杂的视觉-语言任务，我们特别关注其 OCR 能力。为此，我们引入了如下基于 OCR 的预训练：使用 GCP Vision API 检测到的 OCR 文本对 WebLI 数据集 [5] 中的图像进行标注；随后，编码器通过原始基于 JFT 的分类任务与新的基于 OCR 的分类任务（根据 OCR 结果判断给定 token 是否出现在图像中）的混合任务进行进一步预训练。有关视觉组件的更多细节，请参见附录 A。PaLI-X 被设计为以 $n >= 1$ 张图像作为输入（用于少样本和视频理解），涉及单张图像的任务则作为 $n = 1$ 的情况。对于 $n > 1$，每张图像都由 ViT 模块独立处理，从 ViT 输出的 patch 级嵌入会被展平并拼接，以形成视觉输入（见附录 A）。请注意，与单图像情况类似，在视觉嵌入沿时间维度聚合之前，不会在空间维度上进行池化。也就是说，对于一个包含 n 帧、每帧有 k 个 patch 的输入，得到的视觉输入具有 $n * k$ 个 token。

整体模型 编码器-解码器骨干网络初始化自 UL2 [7] 编码器-解码器模型的一个变体，该变体使用 32B 参数。该变体的架构在编码器和解码器中均有 50 层（高于 [7] 中的 32 层），并在与 [7] 类似的文本数据混合集上进行预训练。视觉嵌入在经过一个投影层后，与文本输入的词元嵌入拼接在一起，并输入到编码器-解码器骨干网络中。大多数预训练任务（掩码图像词元任务除外）都从这种多模态输入中预测纯文本输出。模型的文本输入通常由一个提示组成，用于标记任务类型（例如，用于图像描述任务的“*Generate caption in* ⟨lang⟩”），并编码该任务所需的文本输入（例如，用于 VQA 任务的“*Answer in* ⟨lang⟩*: {question}*”）。对于需要 OCR 能力的任务，我们尝试要么仅依赖视觉编码器的文本编码能力，要么可选地将上游 OCR 系统提取的词元作为额外文本输入加入。

Few-shot formulation In the few-shot setting, for a given *target example* the model receives a number of "labeled" examples (in the form of additional ⟨image, text⟩ pairs) that we refer to as *shots/exemplars*. The hypothesis is that information contained in these exemplars provides the model with useful context to generate predictions for the target example. Formally, the input with N shots is a sequence $(t_1, \ldots, t_N, t_T, i_1, \ldots, i_N, i_T)$, where $t_1 : t_N$ and $i_1 : i_N$ are texts and images for the N shots, and t_T and i_T are the text (prompt) and image for the target example. PaLI-X processes this input as follows: all images, including the target one, are first independently processed by the visual encoder, and the resulting patch-level embeddings are flattened and concatenated to form the visual input sequence. After going through a projection layer, they are concatenated with the text embeddings to form the multimodal input sequence used by the encoder. We implement additional optimizations including distributing the exemplars between the encoder and the decoder, and an attention re-weighting mechanism (see Appendix B).

3.2 Pretraining Data and Mixture

The main pretraining data for our model is based on WebLI [5], consisting of roughly one billion images with alt-texts from the web and OCR annotations (using the GCP Vision API), covering over 100 languages. In addition to WebLI ⟨image, text⟩ pairs, we introduce here *Episodic WebLI* data, where each episode corresponds to a set of such pairs. We aim to have each episode contain loosely related images (i.e., they are clustered according to their URL field), so as to encourage attention among examples in an "episode". We find this new dataset (with 75M episodes and around 400M images in total) important for developing the few-shot capabilities of the model.

The pretraining mixture consists of the following data and objectives: (i) span corruption on text-only data (15% of tokens); (ii) split-captioning on WebLI alt-text data [21, 5]; (iii) captioning on CC3M [22] on native and translated alt-text data (over the same 35 languages covered by XM3600 [23]); (iv) split-ocr [24] on WebLI OCR annotations; (v) visual-question-answering objective over ⟨image, question, answer⟩ pairs generated using the VQ^{2}A method [25] over the CC3M training split, over native and translated text (same 35 language pairs); (vi) visual-question-generation objective, using the same pairs as above; (vii) visual-question-answering objective over ⟨image, question, answer⟩ pairs using the Object-Aware method [26] (English only); (viii) captioning on Episodic WebLI examples (target alt-text predicted from the remaining alt-text and images); (ix) visual-question-answering on 4-pair examples (resembling Episodic WebLI and using VQ^{2}A-CC3M pairs), with the answer target conditioned on the other pairs of ⟨image, question, answer⟩ data. (x) pix2struct objective, introduced in [27], targeting page layout and structure using screenshot images paired with DOM-tree representations of html pages. (xi) Captioning on short video data, using the VTP data [10] (using four frames per video). (xii) object-detection objective on WebLI data, whereby an OWL-ViT model [28] (L/14) is used to annotate WebLI images, resulting in hundreds of pseudo object labels and bounding boxes per image. (xiii) image-token prediction objective, whereby we tokenize WebLI images (256×256 resolution) using a ViT-VQGAN [29] model with patch size 16×16 (256 tokens per image); this objective is framed as a 2D masked-token task (i.e., fill-in the missing grid pieces, with the corresponding image pixels also masked). Note that the image-token prediction objective is added mainly as a condition to check whether it adversarially impacts the performance on language-output tasks; our ablation experiments show that is does not.

3.3 Training Stages

Our model is trained in two stages. In stage 1, the visual encoder (after mixed-objective training) is kept frozen, while the rest of the parameters are trained on a total of 2.2B examples at the base resolution 224×224 (native to ViT-22B), using the entire mixture. In stage 2, it continues training using only the OCR-related objectives (pix2struct and split-ocr) plus the object detection objective; this is done in several substages, during which image resolution is gradually increased to 448×448, 672×672 and finally 756×756.

小样本设定 在小样本设置中，对于给定的 *target example*，模型会接收若干“带标签”的示例（以额外的 ⟨图像、文本⟩ 对的形式），我们将其称为 *shots*/*exemplars*。其假设是，这些样例中包含的信息能够为模型提供有用的上下文，以生成目标示例的预测。形式上，包含 N 个样本的输入是一个序列（$t_1, \ldots, t_N, t_T, i_1, \ldots, i_N, i_T$），其中 $t_1 : t_N$ 和 $i_1 : i_N$ 分别是 N 个样本的文本和图像，而 t_T 和 i_T 是目标示例的文本（提示）和图像。PaLI-X 按如下方式处理该输入：所有图像（包括目标图像）首先由视觉编码器独立处理，所得的补丁级嵌入被展平并拼接，以形成视觉输入序列。经过一个投影层后，它们与文本嵌入拼接，形成编码器使用的多模态输入序列。我们实现了额外的优化，包括在编码器和解码器之间分配样例，以及一种注意力重加权机制（见附录 B）。

3.2 预训练数据和混合

我们模型的主要预训练数据基于 WebLI [5]，包含约十亿张来自网络且带有替代文本的图像以及 OCR 注释（使用 GCP Vision API），覆盖 100 多种语言。除了 WebLI ⟨图像、文本⟩对之外，我们在此引入 *Episodic WebLI* 数据，其中每个 episode 对应于一组这样的配对。我们的目标是让每个 episode 包含松散相关的图像（即根据其 URL 字段进行聚类），从而鼓励在一个“episode”中的样例之间进行注意力交互。我们发现这个新数据集（包含 7500 万个 episode，总计约 4 亿张图像）对于发展模型的少样本能力非常重要。

预训练混合数据由以下数据和目标组成：(i) 仅文本数据上的片段损坏（15% 的 token）；(ii) WebLI alt-text 数据上的分割式描述生成 [21, 5]；(iii) CC3M [22] 上基于原生和翻译后的 alt-text 数据的描述生成（覆盖 XM3600 [23] 所涵盖的相同 35 种语言）；(iv) WebLI OCR 标注上的 split-ocr [24]；(v) 在 CC3M 训练划分上，使用 VQ^{2}A 方法 [25] 生成的 ⟨image, question, answer⟩ 三元组上的视觉问答目标，涵盖原生和翻译文本（相同的 35 个语言对）；(vi) 使用上述相同三元组的视觉问题生成目标；(vii) 使用 Object-Aware 方法 [26]（仅英语）生成的 ⟨image, question, answer⟩ 三元组上的视觉问答目标；(viii) Episodic WebLI 示例上的描述生成（根据其余 alt-text 和图像预测目标 alt-text）；(ix) 4 对示例上的视觉问答（类似于 Episodic WebLI，并使用 VQ^{2}A-CC3M 三元组），其中答案目标以其他 ⟨image, question, answer⟩ 数据对为条件；(x) [27] 中引入的 pix2struct 目标，使用与 html 页面 DOM 树表示配对的截图图像来针对页面布局和结构进行建模；(xi) 短视频数据上的描述生成，使用 VTP 数据 [10]（每个视频使用四帧）；(xii) WebLI 数据上的目标检测目标，其中使用 OWL-ViT 模型 [28]（L/14）对 WebLI 图像进行标注，从而为每张图像生成数百个伪对象标签和边界框；(xiii) 图像 token 预测目标，其中我们使用 patch 大小为 16×16 的 ViT-VQGAN [29] 模型对 WebLI 图像（256×256 分辨率）进行 token 化（每张图像 256 个 token）；该目标被设定为 2D 掩码 token 任务（即填补缺失的网格块，同时对应的图像像素也被掩码）。请注意，加入图像 token 预测目标主要是作为一种条件，用于检查它是否会对语言输出任务的性能产生对抗性影响；我们的消融实验表明它不会。

3.3 训练阶段

我们的模型分两个阶段进行训练。在第 1 阶段，视觉编码器（经过混合目标训练后）保持冻结，其余参数则使用完整混合数据，在基础分辨率 224×224（ViT-22B 的原生分辨率）下，总共训练 2.2B 个样本。在第 2 阶段，模型继续仅使用 OCR 相关目标（pix2struct 和 split-ocr）以及目标检测目标进行训练；该阶段分为若干子阶段，在此期间图像分辨率逐步提高到 448×448、672×672，最终达到 756×756。

4 Experiments

4.1 Image Captioning and Visual Question Answering

Our results demonstrate that the larger capacity in PaLI-X scales well in both its vision and language components, and it is particularly beneficial for more challenging scene-text and document understanding tasks. Our model outperforms the SOTA on diverse vision-language tasks, with significant margins in some cases.

Benchmark datasets The Image Captioning and VQA benchmarks used for evaluation is summarized in Appendix B, including 6 Image Captioning benchmarks (COCO (Karpathy split [30]), NoCaps [31], TextCaps [32], VizWiz-Cap [33], Screen2Words [34], Widget-Cap [35]) and 13 VQA benchmarks (VQAv2 [36], OKVQA [37], TallyQA [38], TextVQA [39], VizWiz-VQA [40], STVQA [41], OCRVQA [42], InfographicVQA [43], DocVQA [44], AI2D [45] ChartQA [46], OVEN [47], InfoSeek [48]). These tasks span a wide range of visual domains, from natural images, illustrations to documents and user interfaces (UIs). We also include results of multilingual captioning on XM3600 in Appendix B.

4.1.1 Per-task fine-tuning results

Experimental setup We fine-tune PaLI-X with frozen ViT-22B; the learning rate follows a linear decay from initial value 1e-4 for all fine-tuning experiments. See Appendix B for more details.

	COCO	NoCaps		VQAv2		OKVQA	TallyQA	
Model	Karp.-test	val	test	test-dev	test-std	val	simple	complex
GIT2 [9] (5.1B)	145.0	126.9	**124.8**	81.74	81.92	-	-	-
Flamingo [10] (80B)	138.1	-	-	82.0	82.1	57.8*	-	-
BEiT-3 [49] (1.9B)	147.6	-	-	84.2	84.0	-	-	-
PaLM-E [50] (562B)	138.7	-	-	80.0	-	**66.1**	-	-
MoVie [51]	-	-	-	69.26	-	-	74.9	56.8
PaLI [5](17B)	149.1	**127.0**	124.4	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	126.3	124.3	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**

Table 1: Results on COCO Captions (Karpathy split), NoCaps, VQAv2 [36], OKVQA [37], and TallyQA [38] with end-to-end modeling without OCR pipeline input ("simple" and "complex" are test subsplits).

Model	Text Caps	VizWiz Cap	Text VQA	VizWiz VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	AI2D	Chart QA	Screen2 Words	Widget Cap	OVEN	Info Seek
***with** OCR pipeline input*														
SoTA	160.4 [5]	124.7 [5]	73.67 [52]	73.3 [5]	79.9 [5]	67.5 [53]	47.4 [54]	84.7 [54]	38.5 [45]	45.5 [46]	- -	- -	- -	- -
PaLI-X	**163.7**	**125.7**	**80.78**	**74.6**	**84.5**	**77.3**	**54.8**	**86.8**	**81.4**	**72.3**	-	-	-	-
***without** OCR pipeline input*														
SoTA	145.0 [9]	120.8 [9]	67.27 [9]	70.7 [5]	75.8 [9]	71.3 [27]	40.0 [27]	76.6 [27]	42.1 [27]	70.5 [8]	109.4 [27]	141.8 [20]	20.0 [47]	17.7 [48]
PaLI-X	**147.0**	**122.7**	**71.44**	**70.9**	**79.9**	**75.0**	**49.2**	**80.0**	**81.2**	**70.9**	**127.9**	**153.0**	**23.1**	**21.8**

Table 2: Results on benchmarks more focused on text understanding capabilities. For OVEN [47] & InfoSeek [48], we follow the proposed 224×224 resolution settings for fair comparison.

First, we present benchmarks results for the condition where external OCR systems are not used (Table 1, see Appendix B for an extended table.). The trend is that PaLI-X matches or improves SoTA results on these benchmarks, with a particularly significant improvement on the TallyQA benchmark over MoVie [51] (specialized counting model), at +11.1 for simple counting questions (e.g., "how many giraffes") and +18.8 for complex counting questions (e.g., "how many giraffes are drinking water"); there are significant improvements over PaLI [5] as well, indicating that scale plays an important role in the ability of such models to perform counting tasks. We additionally note the state-of-the-art result on VQAv2 at 86.1 accuracy, achieved with an open-vocabulary generative

4 实验

4.1 图像描述与视觉问答

我们的结果表明，PaLI-X 更大的容量在其视觉和语言组件上都具有良好的扩展性，并且对更具挑战性的场景文本和文档理解任务尤其有益。我们的模型在多种视觉-语言任务上超越了当前最佳水平（SOTA），在某些情况下具有显著优势。

基准数据集 用于评估的图像描述和 VQA 基准汇总见附录 B，其中包括 6 个图像描述基准（COCO（Karpathy 划分 [30]）、NoCaps [31]、TextCaps [32]、VizWiz-Cap [33]、Screen2Words [34]、Widget-Cap [35]）和 13 个 VQA 基准（VQAv2 [36]、OKVQA [37]、TallyQA [38]、TextVQA [39]、VizWiz-VQA [40]、STVQA [41]、OCRVQA [42]、InfographicVQA [43]、DocVQA [44]、AI2D [45]、ChartQA [46]、OVEN [47]、InfoSeek [48]）。这些任务涵盖了广泛的视觉领域，从自然图像、插图到文档和用户界面（UI）。我们还在附录 B 中包含了 XM3600 上多语言图像描述的结果。

4.1.1 按任务微调结果

实验设置 我们使用冻结的 ViT-22B 对 PaLI-X 进行微调；在所有微调实验中，学习率都从初始值 1e-4 线性衰减。更多详细信息见附录 B。

	COCO	NoCaps		VQAv2		OKVQA	TallyQA	
Model	Karp.-test	val	test	test-dev	test-std	val	simple	complex
GIT2 [9] (5.1B)	145.0	126.9	**124.8**	81.74	81.92	-	-	-
Flamingo [10] (80B)	138.1	-	-	82.0	82.1	57.8*	-	-
BEiT-3 [49] (1.9B)	147.6	-	-	84.2	84.0	-	-	-
PaLM-E [50] (562B)	138.7	-	-	80.0	-	**66.1**	-	-
MoVie [51]	-	-	-	69.26	-	-	74.9	56.8
PaLI [5](17B)	149.1	**127.0**	124.4	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	126.3	124.3	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**

表 1：在 COCO Captions（Karpathy 划分）、NoCaps、VQAv2 [36]、OKVQA [37] 和 TallyQA [38] 上的结果，采用无 OCR 流水线输入的端到端建模（“simple”和“complex”为测试子划分）。

Model	Text Caps	VizWiz Cap	Text VQA	VizWiz VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	AI2D	Chart QA	Screen2 Words	Widget Cap	OVEN	Info Seek
***with** OCR pipeline input*														
SoTA	160.4 [5]	124.7 [5]	73.67 [52]	73.3 [5]	79.9 [5]	67.5 [53]	47.4 [54]	84.7 [54]	38.5 [45]	45.5 [46]	- -	- -	- -	- -
PaLI-X	**163.7**	**125.7**	**80.78**	**74.6**	**84.5**	**77.3**	**54.8**	**86.8**	**81.4**	**72.3**	-	-	-	-
***without** OCR pipeline input*														
SoTA	145.0 [9]	120.8 [9]	67.27 [9]	70.7 [5]	75.8 [9]	71.3 [27]	40.0 [27]	76.6 [27]	42.1 [27]	70.5 [8]	109.4 [27]	141.8 [20]	20.0 [47]	17.7 [48]
PaLI-X	**147.0**	**122.7**	**71.44**	**70.9**	**79.9**	**75.0**	**49.2**	**80.0**	**81.2**	**70.9**	**127.9**	**153.0**	**23.1**	**21.8**

表 2：在更侧重文本理解能力的基准测试上的结果。对于 OVEN [47] 和 InfoSeek [48]，我们遵循所提出的 224×224 分辨率设置，以进行公平比较。

首先，我们展示了未使用外部 OCR 系统条件下的基准测试结果（表 1，扩展表见附录 B）。总体趋势是，PaLI-X 在这些基准上达到或超越了 SoTA 结果，尤其是在 TallyQA 基准上相较于 MoVie [51]（专门的计数模型）有显著提升：对于简单计数问题（例如“有多少只长颈鹿”）提升了 +11.1，对于复杂计数问题（例如“有多少只长颈鹿在喝水”）提升了 +18.8；相较于 PaLI [5] 也有显著提升，这表明规模在此类模型执行计数任务的能力中起着重要作用。我们还注意到，在 VQAv2 上达到了 86.1 准确率的最先进结果，这是通过开放词汇生成式

approach, and the performance on OKVQA at 66.1 accuracy, matching the much-larger PaLM-E [50] model performance.

Next, we examine text-heavy V&L benchmarks, for which upstream OCR systems can be used to improve performance. As shown in Table 2, PaLI-X improves SoTA for all Captioning and VQA benchmarks across the board, either without or with additional OCR input (using GCP Vision API). For instance, a significant jump of +42.9 points is observed on AI2D[3], a multiple-choice benchmark where choices are provided along with each question. Being able to have the text choices as input benefits PaLI-X compared with the previous SoTA Pix2Struct [27] which has to render the text on the image, but this does not explain all the improvements. In a question-only configuration (no answer choice present), PaLI-X achieves 46.3 on AI2D, more than 4 points higher than Pix2Struct's result.

In general, having access to OCR texts extracted by an external OCR pipeline boosts performance. Still, for several benchmarks (e.g., AI2D, ChartQA, OCRVQA and Widget-Cap), PaLI-X's end-to-end performance when using its intrinsic OCR capability is close to that leveraging additional OCR input. A common feature for these benchmarks is that they have well-oriented text – diagrams, charts, book covers or user interfaces, with reasonably large font size at 756×756 resolution. For tasks involving scene text in natural images (TextCaps, TextVQA, STVQA) or very high density of small texts (DocVQA, InfoVQA), results still highlight clear benefits when utilizing an external OCR model.

4.1.2 Multitask Fine-tuning

We simultaneously fine-tune and evaluate the pretrained checkpoints on multiple benchmarks belonging to the same category. We deduplicated every training set over the test sets of every task in the mixture to prevent the leakage of any test-set examples into the mixed training set. This is useful as it leads to a single fine-tuned model that performs all the tasks, rather than having to fine-tune each task separately. We performed such multitask fine-tuning on all Image Captioning benchmarks and most VQA benchmarks, respectively.

Table 3 shows the multitask fine-tuning result for captioning tasks. The performance on COCO is slightly decreased in the multitask setting, which is likely a result of this task needing longer training to converge. For Screen2Words, having the smallest train and dev/test sets could be responsible for the performance fluctuation. Notably, VizWiz-Cap and Widget-Cap shows improved performance from multitask fine-tuning. Overall, the average performance decreases by 1.4 points (0.2 excluding Screen2Words) with multitask fine-tuning, while offering the clear advantage of having a single checkpoint to perform all these tasks. Appendix B shows similar results for VQA tasks. We consider this outcome a positive result that establishes the on-par performance between multitask fine-tuning and single-task fine-tuning for diverse benchmarks, in contrast with previous work which argued a gap between single-task and multitask fine-tuning [19], or demonstrated little gap over benchmarks from the same domain [20].

Method	COCO	NoCaps	Text Caps	VizWiz Cap	Screen2 Words	Widget Cap	Avg.
Split	Karp.-test	val	val	test-dev	test	test	-
SOTA (Single-task FT)	149.1	**127.0**	148.6	119.4	109.4	136.7	
PaLI-X Single-task FT	**149.2**	126.3	150.8	123.1	**127.9**	153.2	-
PaLI-X Multitask FT	147.3	125.6	**154.6**	**124.2**	120.6	**153.7**	-
Multitask (+/-)	-1.9	-0.7	+3.8	+1.1	-7.3*	+0.5	-1.4 (-0.2 w/o "*")

Table 3: Scores from multitask fine-tuning compared with those from single-task fine-tuning for Image Captioning. Validation or test-dev set numbers are reported for some tasks.

4.1.3 Few-shot Evaluation

We fine-tuned the PaLI-X model on a mixture of few-shot tasks. The few-shot mixture contains Episodic mixtures, (Non-Episodic) Webli and (Non-Episodic) CC3M data. Note that all of these datasets were already used in previous stages of training, but with lower mixture proportions. During

[3]As with all the other benchmarks, our training examples are carefully deduped to exclude images occurring in these benchmarks, including AI2D. Such results, therefore, are *not* attributable to train-test data leakage.

方法，并且在 OKVQA 上达到 66.1 的准确率，与规模大得多的 PaLM-E [50] 模型性能相当。

接下来，我们考察文本密集型 V&L 基准，在这些基准中，可以使用上游 OCR 系统来提升性能。如表 2 所示，无论是否使用额外的 OCR 输入（使用 GCP Vision API），PaLI-X 都在所有图像描述和 VQA 基准上全面提升了 SoTA。例如，在 AI2D[3] 上观察到 +42.9 分的显著提升；AI2D 是一个多项选择基准，每个问题都会同时提供选项。能够将文本选项作为输入，相比此前的 SoTA Pix2Struct [27] 为 PaLI-X 带来了优势，因为 Pix2Struct 必须将文本渲染到图像上，但这并不能解释全部提升。在仅问题配置（不提供答案选项）下，PaLI-X 在 AI2D 上取得了 46.3 的成绩，比 Pix2Struct 的结果高出 4 分以上。

一般来说，使用由外部 OCR 流水线提取的 OCR 文本可以提升性能。不过，在若干基准测试（例如 AI2D、ChartQA、OCRVQA 和 Widget-Cap）中，PaLI-X 利用其内置 OCR 能力时的端到端性能，接近于使用额外 OCR 输入时的性能。这些基准测试的一个共同特征是文本方向规整——包括图表、统计图、书籍封面或用户界面，并且在 756×756 分辨率下字体大小较为合理。对于涉及自然图像中场景文本的任务（TextCaps、TextVQA、STVQA）或小文本密度非常高的任务（DocVQA、InfoVQA），结果仍然清楚表明，使用外部 OCR 模型能带来明显收益。

4.1.2 多任务微调

我们同时在属于同一类别的多个基准上对预训练检查点进行微调和评估。我们针对混合中每个任务的测试集，对每个训练集进行了去重，以防止任何测试集样本泄漏到混合训练集中。这很有用，因为它会得到一个能够执行所有任务的单一微调模型，而无需分别对每个任务进行微调。我们分别在所有图像描述基准和大多数 VQA 基准上进行了这种多任务微调。

表 3 展示了字幕生成任务的多任务微调结果。在多任务设置下，COCO 上的性能略有下降，这可能是因为该任务需要更长的训练时间才能收敛。对于 Screen2Words，其训练集和开发/测试集规模最小，可能是性能波动的原因。值得注意的是，VizWiz-Cap 和 Widget-Cap 通过多任务微调取得了性能提升。总体而言，多任务微调使平均性能下降了 1.4 个百分点（不包括 Screen2Words 时为 0.2 个百分点），同时带来了一个明显优势：只需单个检查点即可执行所有这些任务。附录 B 展示了 VQA 任务的类似结果。我们认为这一结果是积极的，它证明了在多样化基准上，多任务微调与单任务微调具有相当的性能；这与先前的工作形成对比，后者认为单任务与多任务微调之间存在差距 [19]，或仅在同一领域的基准上显示出很小的差距 [20]。

Method	COCO	NoCaps	Text Caps	VizWiz Cap	Screen2 Words	Widget Cap	Avg.
Split	Karp.-test	val	val	test-dev	test	test	-
SOTA (Single-task FT)	149.1	**127.0**	148.6	119.4	109.4	136.7	
PaLI-X Single-task FT	**149.2**	126.3	150.8	123.1	**127.9**	153.2	-
PaLI-X Multitask FT	147.3	125.6	**154.6**	**124.2**	120.6	**153.7**	-
Multitask (+/-)	-1.9	-0.7	+3.8	+1.1	-7.3*	+0.5	-1.4 (-0.2 w/o "*")

表 3：图像描述任务中，多任务微调得分与单任务微调得分的比较。部分任务报告的是验证集或 test-dev 集的结果。

4.1.3 少样本评估

我们在少样本任务的混合数据上对 PaLI-X 模型进行了微调。该少样本混合数据包含 Episodic 混合数据、（非 Episodic）Webli 和（非 Episodic）CC3M 数据。请注意，所有这些数据集在之前的训练阶段中已经使用过，但混合比例较低。在

[3] As with all the other benchmarks, our training examples are carefully deduped to exclude images occurring in these benchmarks, including AI2D. Such results, therefore, are *not* attributable to train-test data leakage.

pre-training, we only use up to 4 shots, with both encoder and decoder shots (see Appendix B). For fine-tuning, we use up to 8 encoder shots and do not use decoder shots.

We evaluate the few-shot performance on COCO caption (Karpathy test split [30]), and XM3600 [23] datasets. For each task, we first create a "shots pool" with 256 examples that are randomly selected from the task's training set. As the XM3600 benchmark does not come with a training set, we use Google Translate API to enhance the COCO Karpathy training set with captions in the 35 languages represented in XM3600. Then, for each test data point, we randomly pick N shots from the pool as the actual few-shot examples. Following [10], we also evaluate on 2 text-only shots settings where only the textual part of 2 randomly sampled few-shot examples are used.

Table 4 reports the few-shot captioning performance on English and multilingual captioning, as well as few-shot VQA performance on VQAv2. PaLI-X achieves SOTA few-shot results on COCO with both 4 shots and 32 shots; it outperforms previous SOTA by +4.4 CIDEr points for 4-shot, suggesting a strong ability to efficiently gather hints from few examples. We also report few-shot CIDEr scores averaged over 35 languages using XM3600, demonstrating PaLI-X's multilingual capabilities. Meanwhile, although PaLI-X also performs decently on VQAv2, the gap behind the SoTA Flamingo model [10] (which freezes the language backbone) may be the result of losing some of the few-shot text-only QA capability by fine-tuning the language backbone, which supports the hypothesis regarding the tension between few-shot and fine-tuning abilities.

	COCO Captions		XM3600 Cap. (35-lang avg.)		VQAv2	
Method	4 shots	32 shots	4 shots	32 shots	4 shots	32 shots
Prev. SoTA [10]	103.2	113.8	N/A (53.6 w/ fine-tune [5])		**63.1**	**67.6**
PaLI-X	**107.6**	**114.5**	45.1	47.1	56.9	57.1

Table 4: Few-shot performance of the PaLI-X model (multilingual captioning for XM3600).

4.2 Video Captioning and Question Answering

We fine-tune and evaluate the PaLI-X model on 4 video captioning (MSR-VTT [55], VATEX [56], ActivityNet Captions [57], Spoken Moments in Time [58]) and 3 video question answering benchmarks (NExT-QA [59], MSR-VTT-QA [60], ActivityNet-QA [61]). A brief description of each benchmark and clarifications on their usage are provided in Appendix C.

Experimental setup We fine-tune our model (with base resolution 224×224) for each task separately, use the validation split for early stopping, and report performance on the test split. We use a learning rate of 10^{-4} for all tasks, and do not adapt any hyperparameters for specific tasks. Frames are sampled using a fixed temporal stride for each dataset (determined based on the video length distribution in that dataset such that the product of the number of frames and stride is larger than the total number of frames for half of the videos), and we experimented with including up to 8 or 16 frames per video. We did not include pooling over the spatial dimension; embeddings for 16×16 patches per frame are provided as visual input to the multimodal encoder.

Results We report CIDEr score for the video captioning tasks. Video QA tasks are treated as open-ended generation tasks; we report full-string accuracy (for MSR-VTT-QA and ActivityNet-QA) and WUPS metrics (NExT-QA) in [65, 59]. As shown in Table 5, the 16-frames version has an edge over the 8-frame version, sometimes with a significant margin (e.g., close to a 6 point increase in CIDEr score for ActivityNet-Captions). More importantly, while PaLI-X pretraining was dominated by image-text tasks, we were able to achieve new SOTA performance for 5 out of 7 tasks[4], and performed very close to prior SOTA on MSR-VTT-QA (47.1 vs 47.4).

4.3 Image classification

To test image classification capabilities we fine-tuned PaLI-X and models from [5] on ImageNet [66] and evaluated the resulting model on ImageNet-REAL [67] and out-of-distribution

[4]As noted in Table 5, current SOTA on NExT-QA for the open-ended QA task was achieved by Flamingo 32-shot, which had outperformed prior fine-tuning SOTA. To the best of our knowledge, PaLI-X performance on this task does outperform existing published fine-tuning performances, with the caveat that we do not have information on what Flamingo fine-tuning would have achieved on this task.

预训练时，我们最多只使用 4 个样本，同时使用编码器和解码器样本（见附录 B）。微调时，我们最多使用 8 个编码器样本，并且不使用解码器样本。

我们在 COCO caption（Karpathy 测试划分 [30]）和 XM3600 [23] 数据集上评估少样本性能。对于每个任务，我们首先创建一个包含 256 个样例的“样本池”，这些样例从该任务的训练集中随机选取。由于 XM3600 基准没有提供训练集，我们使用 Google Translate API 为 COCO Karpathy 训练集增强了 XM3600 所涵盖的 35 种语言的字幕。然后，对于每个测试数据点，我们从样本池中随机选取 N 个样本作为实际的少样本示例。遵循 [10]，我们还在 2 个纯文本样本设置上进行评估，其中仅使用 2 个随机采样的少样本示例的文本部分。

表 4 报告了英语和多语言图像描述任务上的少样本图像描述性能，以及在 VQAv2 上的少样本 VQA 性能。PaLI-X 在 COCO 上以 4-shot 和 32-shot 均取得了 SOTA 少样本结果；在 4-shot 设置下，它比此前的 SOTA 高出 +4.4 个 CIDEr 点，表明其具有从少量示例中高效获取提示的强大能力。我们还报告了使用 XM3600 在 35 种语言上平均得到的少样本 CIDEr 分数，展示了 PaLI-X 的多语言能力。与此同时，尽管 PaLI-X 在 VQAv2 上的表现也相当不错，但其与 SOTA Flamingo 模型 [10]（该模型冻结了语言主干）之间的差距，可能是由于对语言主干进行微调而损失了部分少样本文本问答能力，这支持了关于少样本能力与微调能力之间存在张力的假设。

	COCO Captions		XM3600 Cap. (35-lang avg.)		VQAv2	
Method	4 shots	32 shots	4 shots	32 shots	4 shots	32 shots
Prev. SoTA [10]	103.2	113.8	N/A (53.6 w/ fine-tune [5])		**63.1**	**67.6**
PaLI-X	**107.6**	**114.5**	45.1	47.1	56.9	57.1

表 4：PaLI-X 模型的少样本性能（XM3600 的多语言图像描述）。

4.2 视频字幕生成与问答

我们在 4 个视频字幕生成基准（MSR-VTT [55]、VATEX [56]、ActivityNet Captions [57]、Spoken Moments in Time [58]）和 3 个视频问答基准（NExT-QA [59]、MSR-VTT-QA [60]、ActivityNet-QA [61]）上对 PaLI-X 模型进行微调和评估。附录 C 提供了每个基准的简要描述以及关于其使用方式的说明。

实验设置 我们针对每个任务分别微调模型（基础分辨率为 224×224），使用验证集划分进行早停，并在测试集划分上报告性能。我们对所有任务使用 10^{-4} 的学习率，并且不针对特定任务调整任何超参数。帧采样对每个数据集使用固定的时间步长（根据该数据集中视频长度分布确定，使得对于一半的视频，帧数与步长的乘积大于总帧数），并且我们实验了每个视频最多包含 8 帧或 16 帧。我们没有在空间维度上进行池化；每帧 16×16 个图像块的嵌入作为视觉输入提供给多模态编码器。

结果 我们报告视频字幕生成任务的 CIDEr 分数。视频问答任务被视为开放式生成任务；我们报告 [65, 59] 中的全字符串准确率（用于 MSR-VTT-QA 和 ActivityNet-QA）以及 WUPS 指标（用于 NExT-QA）。如表 5 所示，16 帧版本优于 8 帧版本，有时优势显著（例如，在 ActivityNet-Captions 上 CIDEr 分数提升接近 6 分）。更重要的是，尽管 PaLI-X 的预训练主要由图像-文本任务主导，我们仍在 7 个任务中的 5 个任务上取得了新的 SOTA 性能[4]，并且在 MSR-VTT-QA 上的表现非常接近此前的 SOTA（47.1 对 47.4）。

4.3 图像分类

为了测试图像分类能力，我们在 ImageNet [66] 上对 PaLI-X 和来自 [5] 的模型进行了微调，并在 ImageNet-REAL [67] 和分布外数据上评估了所得模型。

[4]As noted in Table 5, current SOTA on NExT-QA for the open-ended QA task was achieved by Flamingo 32-shot, which had outperformed prior fine-tuning SOTA. To the best of our knowledge, PaLI-X performance on this task does outperform existing published fine-tuning performances, with the caveat that we do not have information on what Flamingo fine-tuning would have achieved on this task.

Method	MSR-VTT		Activity-Net		VATEX	SMIT	NExT-QA
	Cap. [55]	QA [60]	Cap. [57]	QA [61]	Cap. [56]	Cap. [58]	QA [59]
Prior SOTA	75.9 GIT2 [9]	**47.4** Flamingo [10]	52.5 PDVC [62]	44.7 VINDLU [63]	94.0† GIT2 [9]	28.1‡ MV-GPT [64]	33.5§ Flamingo 32shot [10]
PaLI-X (8fr)	74.6	46.9	49.0	48.4	66.0	42.5	37.0
PaLI-X (16fr)	**76.8**	**47.1**	**54.9**	**49.4**	69.3	**43.5**	**38.3**

Table 5: Results for Video Captioning and Video-QA using 8 frames (8fr) or 16 frames (16fr). †GIT2 uses Self-Critical Sequence Training to directly optimize the CIDEr metric for VATEX. ‡SMIT has not been used for video captioning before, we apply MV-GPT [64] and report results on the test set. §Numbers were obtained using 32-shot; since Flamingo 32-shot outperforms fine-tuning SOTA on this open-ended QA task, they did not conduct further fine-tuning experiments for this task.

datasets: ImageNet-R [68], ImageNet-A [69], ImageNet-Sketch [70], ImageNet-v2 [71]. We used the model from the first training stage (at resolution 224) and the one from the last training stage (at resolution 756). We used the same training hyperparameters for all of runs (selected without any hyperparameter tuning; mode details in Appendix D).

The results can be seen in Table 6. We compare the results to generative model with open vocab – GIT2 [9] (using 384 image resolution), which is the current SOTA for full fine-tuning on ImageNet. PaLI-X achieves SOTA results for generative models on Imagenet, and other datasets. We also performed zero-shot evaluation for PaLI-X and the results can be found in Appendix D.

Model (resolution)	INet [66]	REAL [67]	INet-R [68]	INet-A [69]	INet-Sketch [70]	INet-v2 [71]
GIT2 [9] (384)	**89.22**	-	-	-	-	-
PaLI-17B [5] (224)	86.13	88.84	78.21	50.00	71.21	78.91
PaLI-X (224)	88.22	90.36	77.66	55.97	72.56	81.42
PaLI-X (756)	**89.19**	**90.98**	**80.06**	**72.57**	**73.37**	**83.66**

Table 6: Classification accuracy (top-1) fine-tuned on Imagenet [66].



Credits: Watermelon/Cat; Sarah Pflug (burst), Bowls; ariesandrea (flickr), Wall; Matthew Henry (burst)

Figure 2: Examples demonstrating multilingual, OCR and other capabilities transferred to detection.

4.4 Object Detection

Object detection can be easily formulated in our model as shown in pix2seq [72], The dataset mix used for pre-training is presented in Sec. 3; detection data was included up to and including the stage using resolution 672, after which a separate detection-specific model was fine-tuned on detection data. Before detection-specific tuning, LVIS [73] & COCO labels were removed from all detection training datasets, allowing zero-shot evaluation on LVIS.

Bounding box mean AP on LVIS is shown in Table 7, including zero-shot performance; the detection-tuned model reaches an AP of 31 in general, and 31.4 on rare classes, and about 12 for both in zero-shot. Performance on rare classes was on par with performance on common classes, a difficult

Method	MSR-VTT		Activity-Net		VATEX	SMIT	NExT-QA
	Cap. [55]	QA [60]	Cap. [57]	QA [61]	Cap. [56]	Cap. [58]	QA [59]
Prior SOTA	75.9 GIT2 [9]	**47.4** Flamingo [10]	52.5 PDVC [62]	44.7 VINDLU [63]	94.0† GIT2 [9]	28.1‡ MV-GPT [64]	33.5§ Flamingo 32shot [10]
PaLI-X (8fr)	74.6	46.9	49.0	48.4	66.0	42.5	37.0
PaLI-X (16fr)	**76.8**	**47.1**	**54.9**	**49.4**	69.3	**43.5**	**38.3**

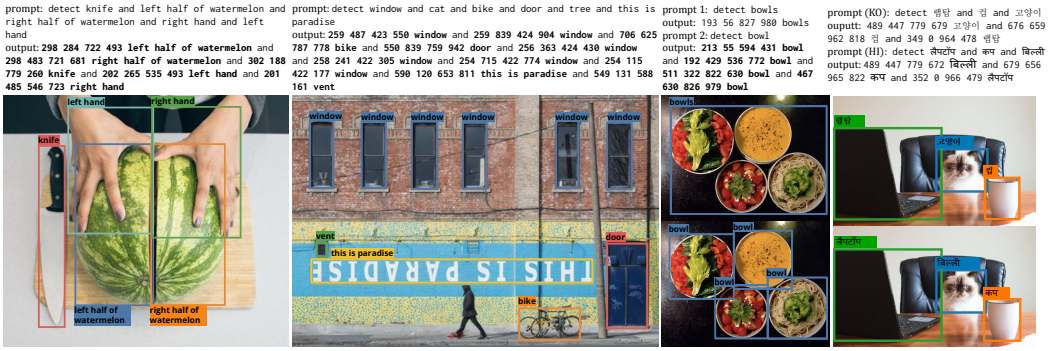
表5：使用8帧（8fr）或16帧（16fr）进行视频字幕生成和视频问答（Video-QA）的结果。† GIT2 使用自我批判序列训练（Self-Critical Sequence Training）来直接优化 VATEX 上的 CIDEr 指标。‡SMIT 此前尚未用于视频字幕生成，我们应用 MV-GPT [64] 并报告测试集上的结果。§这些数值是使用 32-shot 获得的；由于 Flamingo 32-shot 在该开放式问答任务上优于微调的 SOTA，因此他们没有针对此任务进行进一步的微调实验。

数据集：ImageNet-R [68]、ImageNet-A [69]、ImageNet-Sketch [70]、ImageNet-v2 [71]。我们使用了第一训练阶段的模型（分辨率为 224）以及最后训练阶段的模型（分辨率为 756）。我们对所有运行都使用了相同的训练超参数（未进行任何超参数调优；更多细节见附录 D）。

结果见表 6。我们将结果与开放词汇的生成式模型 GIT2 [9]（使用 384 图像分辨率）进行比较，该模型是目前在 ImageNet 上进行全量微调的 SOTA。PaLI-X 在 ImageNet 以及其他数据集上实现了生成式模型的 SOTA 结果。我们还对 PaLI-X 进行了零样本评估，结果见附录 D。

Model (resolution)	INet [66]	REAL [67]	INet-R [68]	INet-A [69]	INet-Sketch [70]	INet-v2 [71]
GIT2 [9] (384)	**89.22**	-	-	-	-	-
PaLI-17B [5] (224)	86.13	88.84	78.21	50.00	71.21	78.91
PaLI-X (224)	88.22	90.36	77.66	55.97	72.56	81.42
PaLI-X (756)	**89.19**	**90.98**	**80.06**	**72.57**	**73.37**	**83.66**

表 6：在 ImageNet [66] 上微调后的分类准确率（top-1）。



Credits: Watermelon/Cat; Sarah Pflug (burst), Bowls; ariesandrea (flickr), Wall; Matthew Henry (burst)

图 2：展示多语言、OCR 及其他能力迁移到检测任务中的示例。

4.4 目标检测

目标检测可以很容易地在我们的模型中表述，如 pix2seq [72] 所示。用于预训练的数据集混合在第 3 节中介绍；检测数据一直被纳入到使用 672 分辨率的阶段（含该阶段），之后在检测数据上对一个单独的检测专用模型进行了微调。在进行检测专用调优之前，已从所有检测训练数据集中移除了 LVIS [73] 和 COCO 标签，从而允许在 LVIS 上进行零样本评估。

LVIS 上的边界框平均 AP 如表 7 所示，包括零样本性能；经过检测调优的模型总体 AP 达到 31，在稀有类别上达到 31.4，而在零样本情况下两者均约为 12。稀有类别上的性能与常见类别上的性能相当，这是一项困难的

feat traditionally accomplished by complicated sampling schedules and augmentations. In our set up, it is directly enabled by PaLI-X's diverse training mix. This could likely be further improved with investment in fine-tuning e.g. using noise-augmentation methods from pix2seq [72], or a further stage of high-resolution, LVIS only training. Qualitatively, we observe emergence of many interesting phenomena enabled by co-training with non-detection tasks; for example, multilingual detection, OCR bounding boxes and longer descriptions, none of which are included in detection training, are often handled well by PaLI-X. Additional results and information can be found in Appendix E.3.

	LVIS AP	LVIS AP_{Rare}
ViLD [74] (tuned on non-rare LVIS)	29.3	26.3
Region-CLIP [75] (tuned on non-rare LVIS)	32.3	22.0
OwLViT-L/16 [28] (tuned on non-rare LVIS)	34.7	25.6
OwLViT-L/16 [28] (with Object365 and VG datasets)	34.6	31.2
PaLI-X (Zeroshot)	12.36	12.16
PaLI-X (Detection-tuned)	30.64	31.42

Table 7: PaLI-X object detection results on LVIS. The diverse pre-training mix enables parity performance between LVIS rare and common classes. Other related approaches are shown for context, but are not directly comparable.

5 Model Fairness, Biases, and Other Potential Issues

Large models, if left unchecked, have the potential to inflict harm on society – such as amplifying biases [76, 77, 78, 79], causing disparities [78, 80, 81], or encoding narrow cultural perspectives [82, 83]. Hence, evaluating PaLI-X for such potential issues is important. We focus our RAI evaluation on three parts: (1) harmful associations, such as toxicity and profanity, (2) demographic parity in the model's output, such as encoding societal stereotypes/biases, and (3) performance disparity across subgroups. This breakdown follows earlier works in the literature, such as [84].

Toxicity / profanity. We estimate the level of toxicity and profanity in the generated captions, including when disaggregated across subgroups. We use the FairFace dataset [85] that comprises of images of people with ground-truth attributes: gender presentation, age and ethnicity. We generate captions and use the Perspective API [86] (threshold > 0.8) to measure toxicity and profanity. Table 8 summarizes the results; we observe a low level of toxicity/profanity across all slices. Tables 9 and 10 provide a detailed breakdown of toxicity/profanity results for all subgroups in FairFace dataset. In Tables 11 and 12, we report similar results in the MIAP [87] dataset, disaggregated by perceived gender and age.

	Gender		Ethnicity			Age			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.14%	0.19%	0.00%	0.13%	0.39%	0.00%	0.17%	0.31%	**0.01%**
Profanity	0.00%	0.02%	0.00%	0.00%	0.05%	0.00%	0.00%	0.03%	**0.00%**

Table 8: Average toxicity/profanity in the captions generated by PaLI-X on FairFace dataset.

Bias / Demographic Parity. We estimate the level of demographic parity (DP) [88] in PaLI-X with respect to gender and occupation. To estimate the level of demographic parity (DP) in the model's output, we feed an image into PaLI-X with the chosen occupation title as a prefix and record the average log-perplexity score of the captions generated by the model. To ensure that any observed parity would likely reflect unintended biases in the model itself as opposed to the evaluation dataset, we use CelebA [89] that contains celebrity images with gender presentation annotation. Our assumption is that many occupations reflecting societal stereotypes, such as secretaries and plumbers, are quite rare in the CelebA dataset so disparities in output may reflect what is encoded in the model itself. The list of occupations is compiled based on [90] and the US job statistics report in [91].

Figure 3 (TOP) summarizes the overall results. First, PaLI-X tends to assign a higher log-perplexity score to women than men across most occupations; i.e. men are predicted to be more likely to hold such occupations. Second, PaLI-X assigns a higher likelihood for a woman to be ('secretary' &

这一成果传统上需要通过复杂的采样计划和数据增强来实现。在我们的设置中，它由 PaLI-X 多样化的训练混合直接促成。通过投入微调，例如使用 pix2seq [72] 中的噪声增强方法，或进一步进行仅针对高分辨率 LVIS 的训练阶段，这一结果很可能还能进一步提升。从定性角度看，我们观察到与非检测任务共同训练带来了许多有趣现象；例如，多语言检测、OCR 边界框和更长的描述，这些都未包含在检测训练中，但 PaLI-X 通常都能很好地处理。更多结果和信息见附录 E.3。

	LVIS AP	LVIS AP_{Rare}
ViLD [74] (tuned on non-rare LVIS)	29.3	26.3
Region-CLIP [75] (tuned on non-rare LVIS)	32.3	22.0
OwLViT-L/16 [28] (tuned on non-rare LVIS)	34.7	25.6
OwLViT-L/16 [28] (with Object365 and VG datasets)	34.6	31.2
PaLI-X (Zeroshot)	12.36	12.16
PaLI-X (Detection-tuned)	30.64	31.42

表 7：PaLI-X 在 LVIS 上的目标检测结果。多样化的预训练组合使 LVIS 稀有类别和常见类别之间达到相当的性能。其他相关方法仅作为背景展示，但不能直接比较。

5 模型公平性、偏见及其他潜在问题

大型模型如果不加以约束，可能会对社会造成伤害——例如放大偏见 [76, 77, 78, 79]、造成差异 [78, 80, 81]，或编码狭隘的文化视角 [82, 83]。因此，评估 PaLI-X 是否存在此类潜在问题十分重要。我们将 RAI 评估重点放在三个方面：(1) 有害关联，例如毒性和亵渎性语言；(2) 模型输出中的人口统计公平性，例如编码社会刻板印象/偏见；以及 (3) 各子群体之间的性能差异。这一划分遵循了文献中的早期工作，例如 [84]。

毒性/脏话。我们估计生成字幕中的毒性和脏话水平，包括按子群体拆分后的情况。我们使用 FairFace 数据集 [85]，该数据集包含带有真实属性标注的人物图像：性别呈现、年龄和族裔。我们生成字幕，并使用 Perspective API [86]（阈值 > 0.8）来衡量毒性和脏话。表 8 总结了结果；我们观察到所有切片中的毒性/脏话水平均较低。表 9 和表 10 提供了 FairFace 数据集中所有子群体的毒性/脏话结果的详细分解。在表 11 和表 12 中，我们报告了 MIAP [87] 数据集上的类似结果，并按感知性别和年龄进行了拆分。

	Gender		Ethnicity			Age			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.14%	0.19%	0.00%	0.13%	0.39%	0.00%	0.17%	0.31%	**0.01%**
Profanity	0.00%	0.02%	0.00%	0.00%	0.05%	0.00%	0.00%	0.03%	**0.00%**

表 8：PaLI-X 在 FairFace 数据集上生成的字幕中的平均毒性/脏话程度。

偏见 / 人口统计均等性。我们评估 PaLI-X 在性别和职业方面的人口统计均等性（DP）[88] 水平。为估计模型输出中的人口统计均等性（DP）水平，我们将一张图像输入 PaLI-X，并以所选职业名称作为前缀，记录模型生成字幕的平均对数困惑度分数。为确保任何观察到的均等性更可能反映模型本身的非预期偏见，而不是评估数据集的偏差，我们使用 CelebA [89]，该数据集包含带有性别呈现标注的名人图像。我们的假设是，许多反映社会刻板印象的职业，例如秘书和水管工，在 CelebA 数据集中相当罕见，因此输出中的差异可能反映的是模型自身编码的内容。职业列表基于 [90] 以及 [91] 中的美国就业统计报告编制。

图 3（顶部）总结了总体结果。首先，在大多数职业中，PaLI-X 倾向于给女性分配比男性更高的对数困惑度分数；也就是说，模型预测男性更有可能从事这些职业。其次，PaLI-X 认为女性成为（“秘书” &

Ethnicity	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
Middle Eastern	64.24%	35.76%	0.00%	94.87%	5.13%	0.00%
Black	59.47%	40.40%	0.13%	92.67%	7.33%	0.00%
Indian	63.86%	36.07%	0.07%	94.39%	5.61%	0.00%
Hispanic	61.09%	38.79%	0.12%	94.45%	5.55%	0.00%
White	62.45%	37.16%	0.39%	92.85%	7.10%	0.05%
Southeast Asian	63.18%	36.61%	0.21%	93.57%	6.43%	0.00%
East Asian	63.15%	36.72%	0.13%	91.55%	8.45%	0.00%

Table 9: Distribution of the predicted toxicity/profanity for the captions generated by PaLI-X on FairFace dataset disaggregated by ethnicity.

Age	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
< 19	58.78%	40.00%	0.22%	89.71%	10.29%	0.00%
20 - 29	63.01%	36.86%	0.12%	93.24%	6.73%	0.03%
30 - 39	63.13%	36.70%	0.17%	95.41%	4.59%	0.00%
40 - 49	63.62%	36.31%	0.07%	95.27%	4.73%	0.00%
50 - 59	65.87%	33.88%	0.25%	96.48%	3.52%	0.00%
60 - 69	65.31%	34.38%	0.31%	95.95%	4.05%	0.00%
> 70	66.10%	33.90%	0.00%	92.37%	7.63%	0.00%

Table 10: Distribution of the predicted toxicity/profanity for the captions generated by PaLI-X on FairFace dataset disaggregated by age.

Perceived Gender	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
Predominantly Feminine	53.98%	45.93%	0.09%	90.55%	9.39%	0.07%
Predominantly Masculine	70.76%	29.17%	0.06%	94.97%	5.01%	0.01%

Table 11: Distribution of the predicted toxicity/profanity for the captions generated by PaLI-X on MIAP dataset disaggregated by perceived gender.

Age Bucket	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
0-2 yrs	28.00%	72.00%	0.00%	69.90%	30.10%	0.00%
3-19 yrs	49.96%	49.96%	0.07%	91.46%	8.54%	0.00%
20-59 yrs	66.27%	33.68%	0.05%	93.42%	6.55%	0.03%
> 60 yrs	65.46%	34.54%	0.00%	96.39%	3.61%	0.00%

Table 12: Distribution of the predicted toxicity/profanity for the captions generated by PaLI-X on MIAP dataset disaggregated by age bucket.

'actor') and a higher likelihood for a man to be ('guard' & 'plumber') at the 95% confidence level. Figure 3 (BOTTOM) displays the corresponding correlations between perceived gender presentation and occupations within the WebLI dataset, where we use the Pearson correlation coefficient by treating each label as a binary random variable and noting that for binary random variables, zero correlation implies full independence. All absolute correlation coefficients in the data are < 0.2 with 99% of them being < 0.1.

Performance Disparity. We present here an evaluation of how well PaLI-X performs across different subgroups using the MIAP [87] dataset. For images containing exactly a single individual, we query PaLI-X with the question: "Is there a person in this image?" and evaluate the accuracy of its response. Note that there are no false positives in this evaluation. Table 13 summarizes the results. We observe that PaLI-X maintains a high accuracy across all subgroups.

Ethnicity	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
Middle Eastern	64.24%	35.76%	0.00%	94.87%	5.13%	0.00%
Black	59.47%	40.40%	0.13%	92.67%	7.33%	0.00%
Indian	63.86%	36.07%	0.07%	94.39%	5.61%	0.00%
Hispanic	61.09%	38.79%	0.12%	94.45%	5.55%	0.00%
White	62.45%	37.16%	0.39%	92.85%	7.10%	0.05%
Southeast Asian	63.18%	36.61%	0.21%	93.57%	6.43%	0.00%
East Asian	63.15%	36.72%	0.13%	91.55%	8.45%	0.00%

表 9：PaLI-X 在 FairFace 数据集上生成的图像说明按族裔细分后的预测毒性/粗俗语言分布。

Age	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
< 19	58.78%	40.00%	0.22%	89.71%	10.29%	0.00%
20 - 29	63.01%	36.86%	0.12%	93.24%	6.73%	0.03%
30 - 39	63.13%	36.70%	0.17%	95.41%	4.59%	0.00%
40 - 49	63.62%	36.31%	0.07%	95.27%	4.73%	0.00%
50 - 59	65.87%	33.88%	0.25%	96.48%	3.52%	0.00%
60 - 69	65.31%	34.38%	0.31%	95.95%	4.05%	0.00%
> 70	66.10%	33.90%	0.00%	92.37%	7.63%	0.00%

表10：PaLI-X 在 FairFace 数据集上生成的字幕按年龄分组后的预测毒性/亵渎性分布。

Perceived Gender	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
Predominantly Feminine	53.98%	45.93%	0.09%	90.55%	9.39%	0.07%
Predominantly Masculine	70.76%	29.17%	0.06%	94.97%	5.01%	0.01%

表 11：PaLI-X 在 MIAP 数据集上生成的字幕中，按感知性别细分的预测毒性/亵渎性分布。

Age Bucket	Toxicity			Profanity		
	< 0.2	$0.2 - 0.8$	> 0.8	< 0.2	$0.2 - 0.8$	> 0.8
0-2 yrs	28.00%	72.00%	0.00%	69.90%	30.10%	0.00%
3-19 yrs	49.96%	49.96%	0.07%	91.46%	8.54%	0.00%
20-59 yrs	66.27%	33.68%	0.05%	93.42%	6.55%	0.03%
> 60 yrs	65.46%	34.54%	0.00%	96.39%	3.61%	0.00%

表 12：按年龄段细分的 MIAP 数据集上 PaLI-X 生成字幕的预测毒性/亵渎性分布。

“演员”），而男性更有可能是（“警卫”和“水管工”），置信水平为 95%。图 3（底部）展示了 WebLI 数据集中感知性别呈现与职业之间的相应相关性，其中我们通过将每个标签视为二元随机变量来使用皮尔逊相关系数，并注意到对于二元随机变量，零相关意味着完全独立。数据中所有相关系数的绝对值均为 < 0.2，其中 99% 为 < 0.1。

性能差异。我们在此使用 MIAP [87] 数据集评估 PaLI-X 在不同子群体上的表现。对于仅包含一个人的图像，我们向 PaLI-X 提问：“这张图像中有人吗？”并评估其回答的准确率。请注意，在此评估中不存在误报。表 13 总结了结果。我们观察到，PaLI-X 在所有子群体上都保持了较高的准确率。

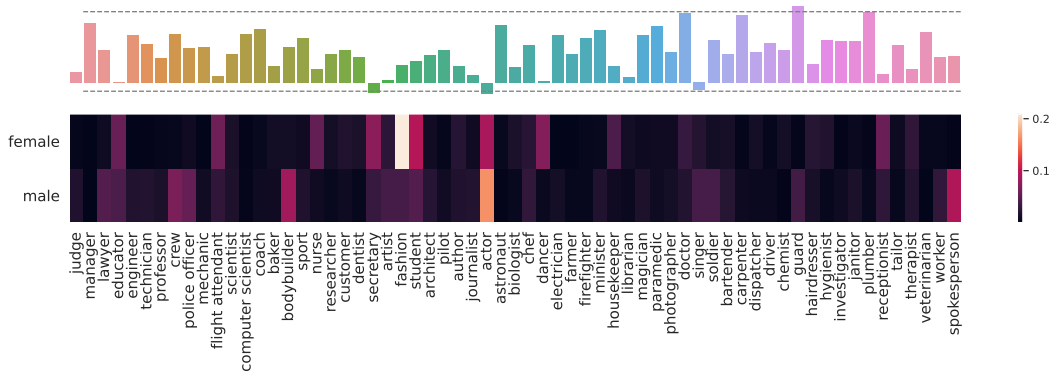


Figure 3: TOP: Level of demographic parity (DP) in PaLI-X's output for CelebA images between women and men. Values close to zero indicate absence of bias. BOTTOM: *Absolute* Pearson correlation coefficients between gender presentation and occupations in WebLI.

Skin Tone	**1** [2]	**2** [871]	**3** [3008]	**4** [522]	**5** [184]	**6** [85]	**7** [54]	**8** [49]	**9** [6]	**10** [1]
	0.00%	0.11%	0.47%	1.53%	0.54%	1.18%	0.00%	0.00%	0.00%	0.00%

Gender	**Predominantly Feminine** [2437]	**Predominantly Masculine** [3544]
	0.53%	0.85%

Age Bucket	**0-2 yrs** [17]	**3-19 yrs** [568]	**20-59 yrs** [4925]	**> 60 yrs** [247]
	0.00%	0.00%	0.77%	0.81%

Table 13: Detection error rate for "person" in PaLI-X using the subset of the MIAP dataset [87] that contain exactly a single individual in the image. PaLI-X maintains a low error rate across all subgroups. Skin tone follows the Monk Skin Tone Scale [92]. Numbers inside square brackets correspond to the size of each bucket.

Limitations. The analysis carried out in this section is necessarily limited, since fairness is a societal concept that cannot be reduced to statistical metrics. We expect RAI evaluations to evolve over time as new issues are detected and reported in the literature and additional datasets become available. Statistical analysis is only a single step and does not substitute for studying the broad and delayed impact of deployed models.

In addition, we rely in some parts on automated tools for inferring attributes, which are not perfectly accurate and can lead to a broad categorization of people that misidentifies real identities. We do not support the creation or application of classifiers for sensitive attributes, such as gender or ethnicity, based on visual indicators and encourage readers to delve into the comprehensive work outlining their potential risks, such as [93, 94], for further insight. Also, while we use perceived gender presentation in our analysis that is provided by the data (i.e. in CelebA and FairFace), we acknowledge that people may express their gendered identities in numerous other ways.

In our evaluation, toxicity is predicted based on the generated captions only. However, without knowing the context of the image, this can introduce false positives.

6 Conclusions

In this work we draw more insights from further scaling vision and language models. We show that the scaling and the improved training recipe results in a model that substantially outperforms previous state-of-the-art models, leads to emergent behaviors and identifies further margins for improvements. In particular, we report that the model achieves significant improvements at document, chart, and infographic understanding, captioning, visual question answering, counting, and performs well on few-shot (in-context) captioning, video captioning and question-answering, and object detection.

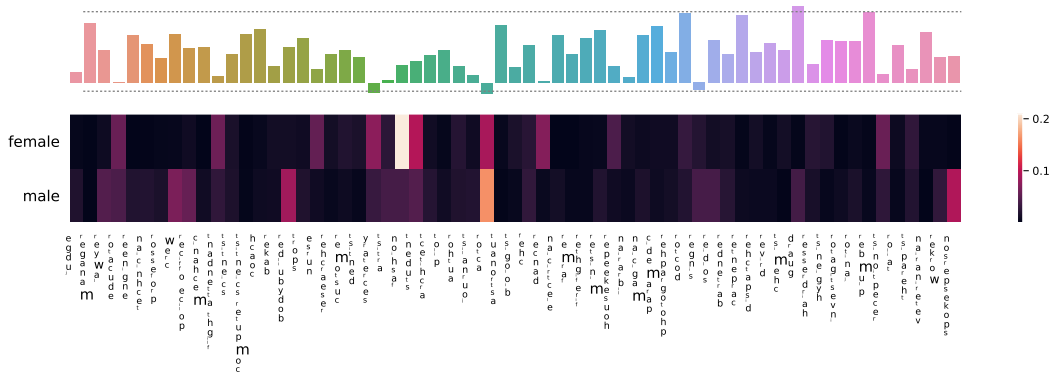


图 3：上图：PaLI-X 对 CelebA 图像的输出中女性与男性之间的人口统计均等性（DP）水平。数值接近零表示不存在偏见。下图：WebLI 中性别呈现与职业之间的 *Absolute* Pearson 相关系数。

Skin Tone	**1** [2]	**2** [871]	**3** [3008]	**4** [522]	**5** [184]	**6** [85]	**7** [54]	**8** [49]	**9** [6]	**10** [1]
	0.00%	0.11%	0.47%	1.53%	0.54%	1.18%	0.00%	0.00%	0.00%	0.00%

Gender	**Predominantly Feminine** [2437]	**Predominantly Masculine** [3544]
	0.53%	0.85%

Age Bucket	**0-2 yrs** [17]	**3-19 yrs** [568]	**20-59 yrs** [4925]	**> 60 yrs** [247]
	0.00%	0.00%	0.77%	0.81%

表 13：PaLI-X 在 MIAP 数据集 [87] 子集上针对“person”的检测错误率，该子集中的图像恰好只包含一个个体。PaLI-X 在所有子群体中均保持较低错误率。肤色遵循 Monk 肤色量表 [92]。方括号内的数字对应每个桶的大小。

局限性。本节所进行的分析必然是有限的，因为公平性是一个社会概念，无法被简化为统计指标。我们预计，随着文献中发现并报告新的问题以及更多数据集的可用，RAI 评估将随着时间的推移而演进。统计分析只是其中的一个步骤，不能替代对已部署模型所产生的广泛且滞后影响的研究。

此外，我们在某些部分依赖自动化工具来推断属性，而这些工具并非完全准确，可能会导致对人群进行宽泛分类，从而错误识别真实身份。我们不支持基于视觉指标创建或应用用于敏感属性（如性别或族裔）的分类器，并鼓励读者深入阅读阐述其潜在风险的综合性工作，如 [93, 94]，以获得进一步了解。此外，尽管我们在分析中使用了由数据提供的感知性别呈现（即 CelebA 和 FairFace 中的数据），但我们承认，人们可能以许多其他方式表达其性别身份。

在我们的评估中，毒性仅基于生成的字幕进行预测。然而，如果不了解图像的上下文，这可能会导致误报。

6 结论

在这项工作中，我们通过进一步扩展视觉和语言模型获得了更多见解。我们表明，规模扩展和改进的训练方案带来了一个显著超越以往最先进模型的模型，产生了涌现行为，并指出了进一步改进的空间。具体而言，我们报告该模型在文档、图表和信息图理解、图像描述、视觉问答、计数方面取得了显著提升，并且在少样本（上下文内）图像描述、视频描述和问答以及目标检测方面表现良好。

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A Additional Model Details and Examples

A.1 PaLI-X Architecture Illustration

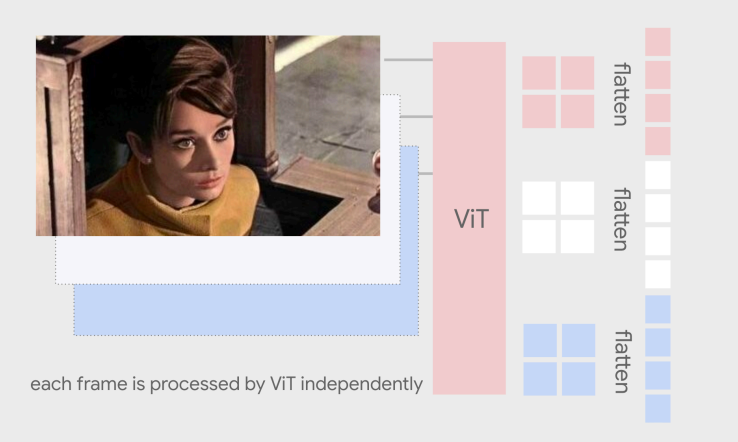


Figure 4: Visual input for videos: each frame is independently processed by ViT; patch embeddings are flattened and concatenated together to form the visual representation. (The example input image is in the public domain).

A.2 Tuning ViT-22B for better OCR capabilities

The vision encoder's ability to understand text is crucial to several downstream tasks and general usability. JFT-based pre-training is insufficient to cover this, and so we tuned ViT-22B on WebLI-OCR data. In order to stay true to the original discriminative classification-based objective used for ViT-22B, we turn OCR into a bag-of-words prediction task. OCR texts are tokenized using the mT5 tokenizer [95] across all languages, and the model is trained to predict whether or not a given token occurs in an image. This is treated as multilabel classification, with an expanded classification head.

In the ablation study shown in Table 22, we confirm that this this extra tuning step indeed has a significant improvement on Scene-Text understanding capabilities, demonstrated by the performance on ST-VQA and TextVQA. Meanwhile, the performance on regular VQA tasks such as those in the VQAv2 benchmark also improves.

A.3 Illustrative PaLI-X Examples

Table 14 shows representative examples of PaLI-X, illustrating improved abilities related to counting (both of the simple and complex variety), in context text-reading capabilities, and spatial awareness.

A 额外的模型细节和示例

A.1 PaLI-X 架构示意图

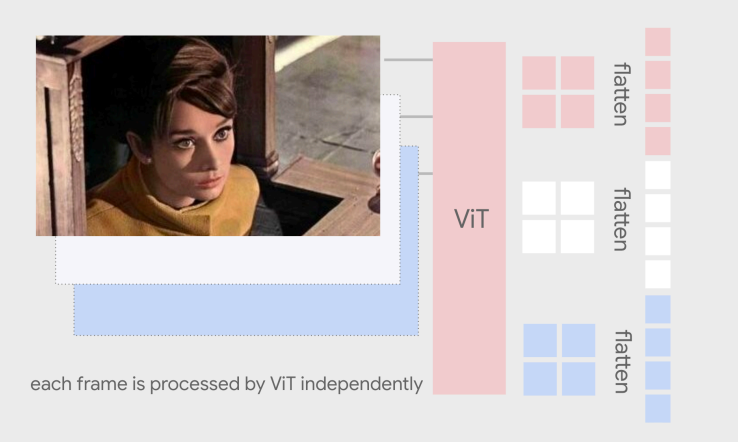


图 4：视频的视觉输入：每一帧都由 ViT 独立处理；图像块嵌入被展平并拼接在一起，以形成视觉表示。（示例输入图像属于公有领域）。

A.2 调优 ViT-22B 以获得更好的 OCR 能力

视觉编码器理解文本的能力对于多个下游任务和整体可用性至关重要。基于 JFT 的预训练不足以覆盖这一能力，因此我们在 WebLI-OCR 数据上对 ViT-22B 进行了调优。为了保持与 ViT-22B 最初使用的基于判别式分类的目标一致，我们将 OCR 转化为词袋预测任务。OCR 文本使用 mT5 分词器 [95] 对所有语言进行分词，并训练模型预测给定的词元是否出现在图像中。这被视为多标签分类，并使用扩展的分类头。

在表 22 所示的消融研究中，我们确认这一额外的调优步骤确实显著提升了场景文本理解能力，这一点由 ST-VQA 和 TextVQA 上的表现所证明。与此同时，在 VQAv2 基准等常规 VQA 任务上的表现也有所提升。

A.3 PaLI-X 示例说明

表14展示了 PaLI-X 的代表性示例，说明其在计数（包括简单和复杂类型）、上下文文本阅读能力以及空间感知方面的能力有所提升。



Image Credit: Wikimedia Commons [CC BY-SA 4.0]

Q: how many coins are there?
A: 12

Q: how many one-dollar coins are there?
A: 2

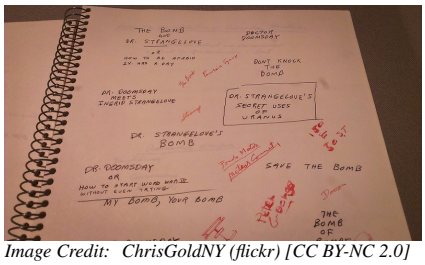


Image Credit: ChrisGoldNY (flickr) [CC BY-NC 2.0]

Q: what is written inside the box?
A: dr. strangelove's secret uses of uranus

Q: what is written on the top-left corner of the page?
A: the bomb and dr. strangelove

Q: what is written on the top-right corner of the page?
A: doctor doomsday



Public Domain Image

Captioning:
a movie poster for rebecca starring laurence olivier and joan fontaine.

Q: who directed this movie?
A: alfred hitchcock

Q: who produced this movie?
A: david o. seznick

Table 14: Examples of counting, text reading capabilities with context and spatial awareness. Results are generated by the multi-task-finetuned models using the model's inherent OCR capabilities (i.e., without the use of an external OCR system).

B Additional results: Image Captioning and VQA

B.1 Information of Downstream Image Benchmarks

Table 15 summarizes the Image Captioning and VQA benchmarks. For benchmarks modeled only end-to-end without OCR pipeline input (Table 1 and Table 16), fine-tuning is performed with resolution 672×672. For Scene-Text and Document Understanding tasks presented in Table 2, fine-tuning is performed with resolution 756×756.

B.2 Extended Tables of Image Benchmarks

An extended table of results on some Image Benchmarks is shown as Table 16.



Image Credit: Wikimedia Commons [CC BY-SA 4.0]

Q: how many coins are there?
A: 12

Q: how many one-dollar coins are there?
A: 2

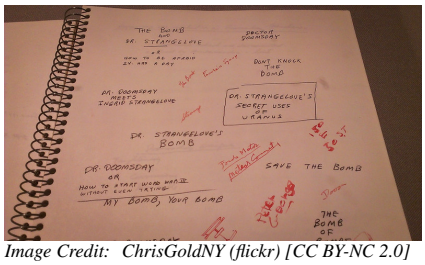


Image Credit: ChrisGoldNY (flickr) [CC BY-NC 2.0]

Q: what is written inside the box?
A: dr. strangelove's secret uses of uranus

Q: what is written on the top-left corner of the page?
A: the bomb and dr. strangelove

Q: what is written on the top-right corner of the page?
A: doctor doomsday



Public Domain Image

Captioning:
a movie poster for rebecca starring laurence olivier and joan fontaine.

Q: who directed this movie?
A: alfred hitchcock

Q: who produced this movie?
A: david o. seznick

表14：具有上下文和空间感知能力的计数、文本阅读能力示例。结果由多任务微调模型利用模型自身固有的 OCR 能力生成（即未使用外部 OCR 系统）。

B 附加结果：图像描述和 VQA

B.1 下游图像基准测试信息

表15总结了图像描述和VQA基准。对于仅以端到端方式建模且不输入OCR流水线的基准（表1和表16），微调采用672×672的分辨率进行。对于表2中呈现的场景文本和文档理解任务，微调采用756×756的分辨率进行。

B.2 图像基准测试的扩展表格

表 16 展示了一些图像基准测试的扩展结果表。

Benchmark	Visual Domain	Description	Metric
COCO Captions	Natural Images	Captioning of natural images	CIDEr
NoCaps		Captioning of natural images	CIDEr
TextCaps		Captioning of natural images containing text	CIDEr
VizWiz-Cap		Captioning of photos taken by people who are blind	CIDEr
VQAv2		VQA on natural images	VQA accu.
OKVQA		VQA on natural images requiring outside knowledge	VQA accu.
TextVQA		VQA on natural images containing text	VQA accu.
VizWiz-QA		VQA on photos taken by people who are blind	VQA accu.
ST-VQA		VQA on natural images containing text	ANLS
TallyQA		VQA with counting questions	EM
OVEN		VQA on natural images for visual entity recognition	EM
InfoSeek		VQA on natural images for visual info-seeking questions	Relaxed EM
OCR-VQA	Illustrations	VQA on images of book covers	EM
ChartQA		VQA on images of charts	RA
AI2D		VQA on images of scientific diagrams	EM
DocVQA	Documents	VQA on images of scanned documents	ANLS
InfographicsVQA		VQA on images of infographics	ANLS
Screen2Words	UIs	Captioning a UI screen to describe functionality	CIDEr
Widget Captioning		Captioning a UI component on a screen	CIDEr

Table 15: Summary of Image Captioning and VQA benchmarks used for evaluating PaLI-X

	COCO	NoCaps		VQAv2		OKVQA	TallyQA	
Model	Karp.-test	val	test	test-dev	test-std	val	simple	complex
SimVLM	143.3	112.2	110.3	80.03	80.34	-	-	-
CoCa (2.1B)	143.6	122.4	120.6	82.3	82.3	-	-	-
GIT (0.7B)	144.8	125.5	123.4	78.56	78.81	-	-	-
GIT2 (5.1B)	145.0	126.9	**124.8**	81.74	81.92	-	-	-
OFA (0.9B)	145.3	-	-	82.0	82.0	-	-	-
Flamingo (80B)	138.1	-	-	82.0	82.1	57.8*	-	-
BEiT-3 (1.9B)	147.6	-	-	84.2	84.0	-	-	-
PaLM-E (562B)	138.7	-	-	80.0	-	**66.1**	-	-
MoVie	-	-	-	69.26	-	-	74.9	56.8
PaLI (17B)	149.1	**127.0**	124.4	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	126.3	124.3	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**

Table 16: Results on COCO Captions (Karpathy split), NoCaps, VQAv2, OKVQA, and TallyQA with end-to-end modeling without OCR pipeline input. The "simple" and "complex" are test subsplits.

B.3 Multi-lingual Captioning

Multilingual captioning on XM-3600 The Crossmodal-3600 (XM3600) benchmark contains a geo-diverse set of 3600 images with human-annotated reference captions in 36 languages [23]. Table 17 presents multilingual results for both PaLI (current SoTA on XM-3600) and PaLI-X, both finetuned with 224×224 resolution. Overall, PaLI-X improves on the SoTA performance across 5 of the 7 languages we report here (and for 14 of the total 35 languages considered); notably, the performance on English is 4 CIDEr points lower compared to PaLI. The 35-language average CIDEr score is in the same ballpark between PaLI and PaLI-X, with a slight +0.5 advantage for PaLI.

Model	en	fr	hi	iw	ro	th	zh	35-lang avg.
PaLI	**98.1**	75.5	31.3	46.8	35.8	72.1	**36.5**	**53.6**
PaLI-X	94.2	**78.7**	**32.0**	**46.9**	**36.9**	**75.3**	36.1	53.1

Table 17: CIDEr scores on image captioning for the Crossmodal-3600 benchmark for seven diverse languages (English, French, Hindi, Hebrew, Romanian, Thai, and Chinese), as well as the average of the 35 languages covered by the benchmark. Both models are finetuned with 224×224 resolution.

Benchmark	Visual Domain	Description	Metric
COCO Captions	Natural Images	Captioning of natural images	CIDEr
NoCaps		Captioning of natural images	CIDEr
TextCaps		Captioning of natural images containing text	CIDEr
VizWiz-Cap		Captioning of photos taken by people who are blind	CIDEr
VQAv2		VQA on natural images	VQA accu.
OKVQA		VQA on natural images requiring outside knowledge	VQA accu.
TextVQA		VQA on natural images containing text	VQA accu.
VizWiz-QA		VQA on photos taken by people who are blind	VQA accu.
ST-VQA		VQA on natural images containing text	ANLS
TallyQA		VQA with counting questions	EM
OVEN		VQA on natural images for visual entity recognition	EM
InfoSeek		VQA on natural images for visual info-seeking questions	Relaxed EM
OCR-VQA	Illustrations	VQA on images of book covers	EM
ChartQA		VQA on images of charts	RA
AI2D		VQA on images of scientific diagrams	EM
DocVQA	Documents	VQA on images of scanned documents	ANLS
InfographicsVQA		VQA on images of infographics	ANLS
Screen2Words	UIs	Captioning a UI screen to describe functionality	CIDEr
Widget Captioning		Captioning a UI component on a screen	CIDEr

表 15：用于评估 PaLI-X 的图像描述和 VQA 基准汇总

	COCO	NoCaps		VQAv2		OKVQA	TallyQA	
Model	Karp.-test	val	test	test-dev	test-std	val	simple	complex
SimVLM	143.3	112.2	110.3	80.03	80.34	-	-	-
CoCa (2.1B)	143.6	122.4	120.6	82.3	82.3	-	-	-
GIT (0.7B)	144.8	125.5	123.4	78.56	78.81	-	-	-
GIT2 (5.1B)	145.0	126.9	**124.8**	81.74	81.92	-	-	-
OFA (0.9B)	145.3	-	-	82.0	82.0	-	-	-
Flamingo (80B)	138.1	-	-	82.0	82.1	57.8*	-	-
BEiT-3 (1.9B)	147.6	-	-	84.2	84.0	-	-	-
PaLM-E (562B)	138.7	-	-	80.0	-	**66.1**	-	-
MoVie	-	-	-	69.26	-	-	74.9	56.8
PaLI (17B)	149.1	**127.0**	124.4	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	126.3	124.3	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**

表 16：在不使用 OCR 流水线输入的端到端建模下，COCO Captions（Karpathy 划分）、NoCaps、VQAv2、OKVQA 和 TallyQA 上的结果。“simple”和“complex”是测试子划分。

B.3 多语言字幕生成

XM-3600 上的多语言图像描述 Crossmodal-3600（XM3600）基准包含一组地理分布多样的 3600 张图像，并配有 36 种语言的人工标注参考描述 [23]。表 17 展示了 PaLI（当前 XM-3600 上的 SoTA）和 PaLI-X 的多语言结果，二者均以 224×224 分辨率进行微调。总体而言，在我们此处报告的 7 种语言中，PaLI-X 在其中 5 种语言上提升了 SoTA 性能（在所考虑的全部 35 种语言中有 14 种也是如此）；值得注意的是，英语上的性能相比 PaLI 低了 4 个 CIDEr 点。PaLI 和 PaLI-X 的 35 种语言平均 CIDEr 分数处于相近水平，PaLI 略有 +0.5 的优势。

Model	en	fr	hi	iw	ro	th	zh	35-lang avg.
PaLI	**98.1**	75.5	31.3	46.8	35.8	72.1	**36.5**	**53.6**
PaLI-X	94.2	**78.7**	**32.0**	**46.9**	**36.9**	**75.3**	36.1	53.1

表17：在 Crossmodal-3600 基准上，七种不同语言（英语、法语、印地语、希伯来语、罗马尼亚语、泰语和中文）的图像描述 CIDEr 分数，以及该基准涵盖的 35 种语言的平均分。两个模型均以 224×224 分辨率进行微调。

B.4 TallyQA and the emergence of complex counting capability

We present in Table 18 the performance of similar models across a wide range of capacity – from 700M parameters to 55B parameters for PaLI-X. The graphs in Fig. 5 illustrate how simple counting appears to follow a more linear progression as parameter-size increases, while complex counting appears to show emergence somewhere before the datapoint provided by the performance of PaLI 17B. This corresponds to our intution that complex counting is a true multimodal task that requires additional capabilities from a model, in terms of the alignment that is required between the visual information and the prompt specification.

Model	TallyQA simple	TallyQA complex	Weighted average
PaLI (700M)	66.9	55.6	62.4
PaLI (3B)	72.0	56.7	65.9
PaLI (17B)	76.2	65.5	71.9
PaLI-X (55B)	81.3	71.0	77.2

Table 18: Performance on TallyQA splits for simple and complex questions. All models use 224×224 image resolution.

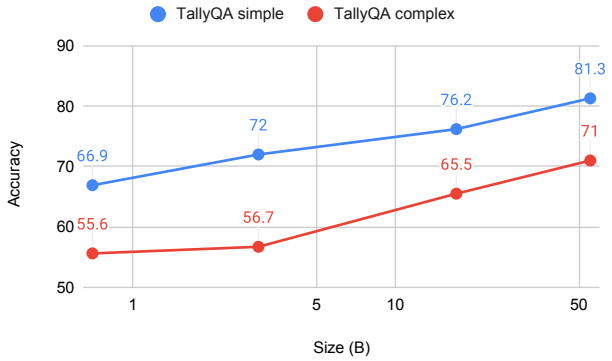


Figure 5: Performance on TallyQA splits for simple and complex using PaLI variants and PaLI-X. All models use 224×224 image resolution. The emergent behavior on complex counting beyond the 3B size is made clear with PaLI-X.

B.5 Details on Few-shot Modeling

B.5.1 Few-shot Formulation

Figure 6 illustrates the network flow of a few shot model. The text and prompt part of each shot is embedded and concatenated as text features for the PaLI-X model. Each shot's images and the target image are independently encoded by the ViT component, and the ViT features are concatenated along the sequence axis as visual features. Conditioned on that sequence, the PaLI-X decoder autoregressively makes the predictions for the target image.

Encoder shot and Decoder shots While images for all few-shot examples and target example are given as input to the model, text information can be provided in different ways. During inference time, all text information related to the few-shot examples is given to the encoder; in the case of a Multi-answer VQA task, for example, this includes both the prompts that contain the questions, and the expected answers. Prompt for the target example is also given to the encoder, and the decoder is tasked with generating an answer for the target example. During training, however, we increase the training efficiency by making the model predict answers for both the target example and selected shots (the *decoder shots*). That is, we partition the N shots in two sets: encoder shots ($N_e > 0$) and decoder shots ($N_d \geq 0$), such that $N_e + N_d \leq N$. We use up to 4 shots in total during pre-training (i.e. $N = 4$), and sample N_e uniformly at random from 1 to N. Text input for encoder shots contain

B.4 TallyQA 与复杂计数能力的涌现

我们在表 18 中展示了类似模型在广泛容量范围内的性能——从 700M 参数到 PaLI-X 的 55B 参数。图 5 中的图表说明，随着参数规模增加，简单计数似乎呈现出更线性的进展，而复杂计数似乎在 PaLI 17B 的性能所提供的数据点之前的某个位置表现出涌现。这与我们的直觉相符：复杂计数是一项真正的多模态任务，需要模型具备额外能力，即视觉信息与提示规范之间所需的对齐能力。

Model	TallyQA simple	TallyQA complex	Weighted average
PaLI (700M)	66.9	55.6	62.4
PaLI (3B)	72.0	56.7	65.9
PaLI (17B)	76.2	65.5	71.9
PaLI-X (55B)	81.3	71.0	77.2

表18：在 TallyQA 简单问题和复杂问题划分上的性能。所有模型均使用 224×224 的图像分辨率。

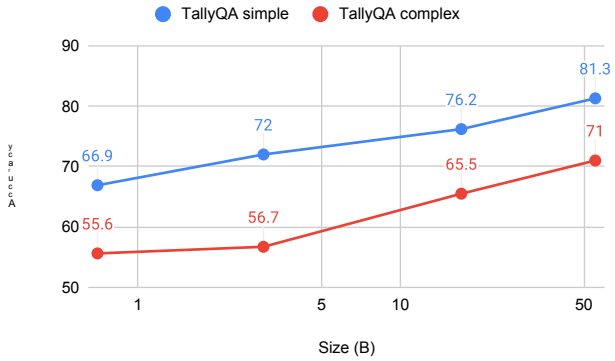


图 5：使用 PaLI 变体和 PaLI-X 在 TallyQA 简单与复杂划分上的性能。所有模型均使用 224×224 的图像分辨率。PaLI-X 清楚地展现了超过 3B 规模后在复杂计数上的涌现行为。

B.5 少样本建模细节

B.5.1 少样本形式化

图 6 展示了少样本模型的网络流程。每个样本的文本和提示部分被嵌入并拼接为 PaLI-X 模型的文本特征。每个样本的图像和目标图像由 ViT 组件独立编码，并且 ViT 特征沿序列轴拼接为视觉特征。在该序列的条件下，PaLI-X 解码器以自回归方式对目标图像进行预测。

编码器样本和解码器样本 虽然所有少样本示例和目标示例的图像都会作为输入提供给模型，但文本信息可以通过不同方式提供。在推理时，与少样本示例相关的所有文本信息都会提供给编码器；例如，在多答案 VQA 任务中，这包括包含问题的提示以及预期答案。目标示例的提示也会提供给编码器，而解码器的任务是为目标示例生成答案。然而，在训练期间，我们通过让模型同时预测目标示例和所选样本（*decoder shots*）的答案来提高训练效率。也就是说，我们将 N 个样本划分为两组：编码器样本（$N_e > 0$）和解码器样本（$N_d \geq 0$），使得 $N_e + N_d \leq N$。我们在预训练期间总共最多使用 4 个样本（即 $N = 4$），并从 1 到 N 中均匀随机采样 N_e。编码器样本的文本输入包含

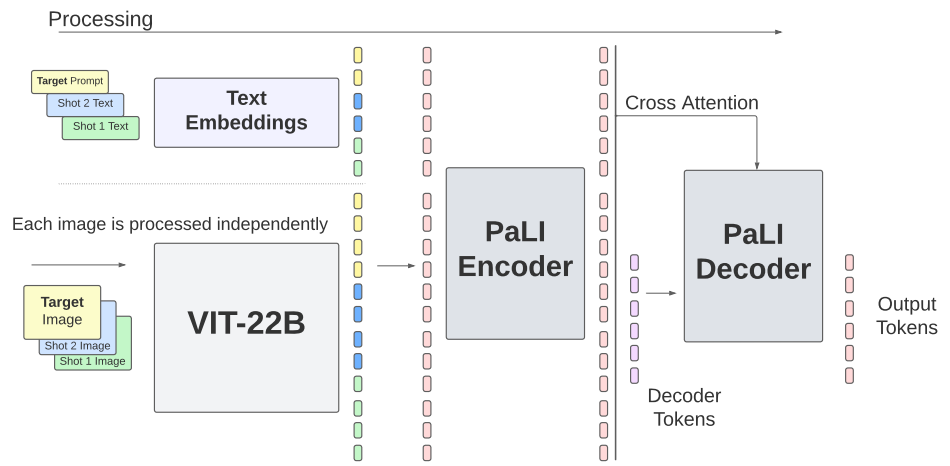


Figure 6: A detailed view on how the few-shot exemplars are fed to the model components.

both prompts and answers. The decoder shots, however, act as if they were target examples: their text input to the encoder contains only the prompt, and the decoder needs to predict answers for the decoder shots in addition to the target example.

Attention re-weighting Increasing the number of shots turned out to be challenging, potentially due to cross-attention to target example input tokens getting diluted by the large number of shots. To address this, we introduce an attention re-weighting mechanism. As shown in Figure 7, we explicitly boost the weights for cross attention between decoder tokens and encoded tokens from the target example (that is, the target image and the target text prompt).

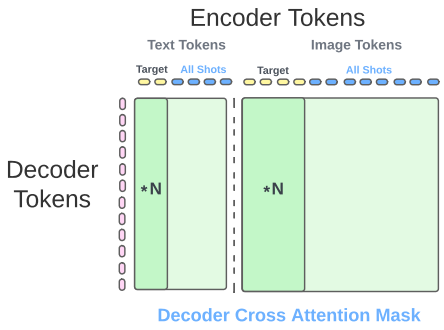


Figure 7: Re-weighted attention with few-shots.

Specifically, if there are N shots in total, when decoding each token we multiply the cross attention weights by N for the target image and text tokens from the encoder outputs. We observe this attention re-weighting technique is especially helpful when we provide the model with many shots (e.g. 32 shots). [96] introduces a technique along similar lines to manipulate attention weights when gathering them from different threads of encoded shots at inference time.

B.5.2 Additional Few-shot Results

Multilingual captioning results Table 19 reports the CIDEr scores for 7 languages and an average over 35 languages to demonstrate PaLI's multilingual captioning capabilities on the XM3600 benchmark in teh few-shot setting. The pre-trained model (no few-shot finetuning) achieves an average score of 22.7. The PaLI-X model achieves an average score of 45.1 for 4 shots and 47.1 for 32 shots. Note that the 32-shot PaLI-X average CIDEr score is only 6 points behind the fully finetuned model, which uses roughly 600k training examples per language (while the few-shot approach does not update the model parameters).

Qualitative results Figure 8 shows 3 examples on few-shot captioning and VQA tasks for qualitative analysis. The first row shows captions for the images using the images' original language,

[5]Equivalent with the Flamingo "0-shot" setting.

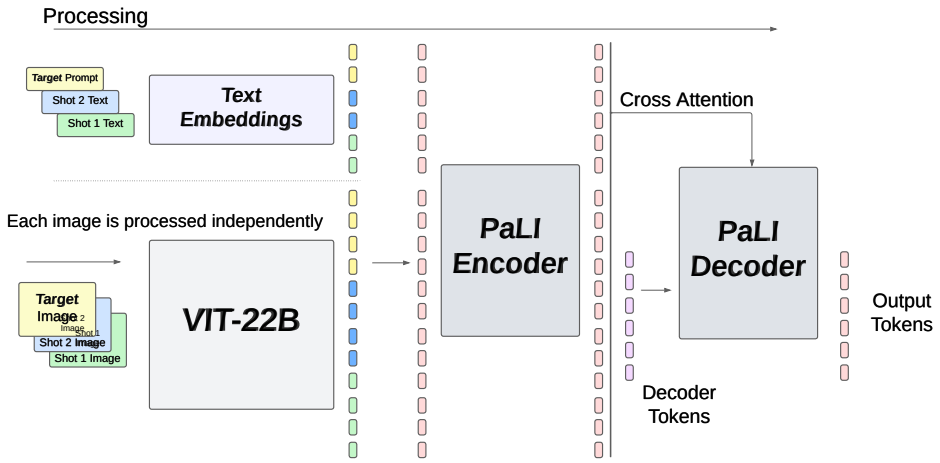


图 6：少样本示例如何被输入到模型组件中的详细视图。

同时包含提示和答案。然而，解码器示例的行为就像目标样例一样：它们输入到编码器的文本只包含提示，而解码器除了需要预测目标样例的答案外，还需要预测解码器示例的答案。

注意力重加权 增加 shots 的数量被证明具有挑战性，这可能是由于对目标示例输入 token 的交叉注意力被大量 shots 稀释了。为了解决这一问题，我们引入了一种注意力重加权机制。如图 7 所示，我们显式提升了解码器 token 与来自目标示例的编码 token（即目标图像和目标文本提示）之间的交叉注意力权重。

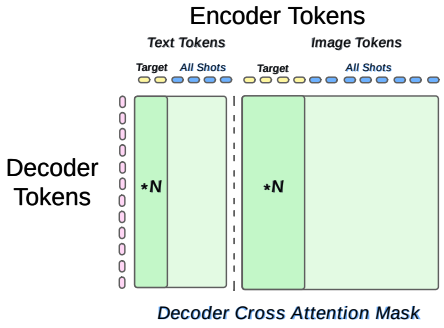


图 7：少样本下的重加权注意力。

具体而言，如果总共有 N 个 shots，那么在解码每个 token 时，我们会将来自编码器输出的目标图像和文本 token 的交叉注意力权重乘以 N。我们观察到，当我们为模型提供大量 shots（例如 32 shots）时，这种注意力重新加权技术尤其有帮助。[96] 引入了一种类似思路的技术，用于在推理时从不同的编码镜头线程中收集注意力权重时对其进行操控。

B.5.2 额外的小样本结果

多语言图像描述结果 表19报告了7种语言的CIDEr分数以及35种语言的平均分，以展示PaLI在XM3600基准上少样本设置下的多语言图像描述能力。预训练模型（未进行少样本微调）的平均分为22.7。PaLI-X模型在4-shot时的平均分为45.1，在32-shot时为47.1。请注意，32-shot PaLI-X的平均CIDEr分数仅比完全微调模型低6分，而完全微调模型每种语言大约使用60万个训练样本（而少样本方法不会更新模型参数）。

定性结果 图8展示了少样本图像描述和VQA任务中用于定性分析的3个示例。第一行展示了使用图像原始语言生成的图像描述，

[5]Equivalent with the Flamingo “0-shot” setting.



Figure 8: Qualitative Results on few-shot captioning (first two rows) and VQA (the last row) tasks.

	Crossmodal-3600 Captioning							
	en	fr	hi	iw	ro	th	zh	35-lang avg.
PaLI-X 0-shot	48.8	25.0	10.5	20.1	13.0	33.3	18.4	22.7
PaLI-X (2 text-only shots[5])	54.5	46.7	12.0	22.2	9.4	40.3	23.7	25.8
PaLI-X 4 shots	77.8	62.5	22.2	38.7	30.2	56.0	27.7	45.1
PaLI-X 32 shots	81.4	66.1	25.6	40.6	32.4	59.4	29.7	47.1
PaLI-X (finetuned)	94.2	78.7	32.0	46.9	36.9	75.3	36.1	53.1

Table 19: Few-shot performance of the PaLI-X model on multilingual captioning tasks.

demonstrating the cross multilingual transfer of the few-shot capability. The second row captions the images with a country's popular food, showing that the few-shot approach can access the model's world knowledge. The last row shows a VQA with an explanation-like scenario where we ask if the technologies in the images are "new". Generally speaking, the shown personal computer was produced more than 40 years ago and could be regarded as old technology considering the fast pace of the current high-tech development. However, the 3 input shots provide the detailed calibration for the concept of "new" and the few-shot model successfully take the context and output "new" with plausible explanation to the very old PC.

B.5.3 Few-shot ablation results

In this section, we present and discuss some ablation results for few-shot we explored in order to inform our final design choices on PaLI-X. Unless otherwise specified, we use a 700M-parameter model with the same encoder-decoder architecture, consisting of a ViT-B/16 vision encoder and a mT5-base encoder-decoder language model.

Pooling vs not pooling image tokens To mitigate the computational burden that arises with many shots, we can pool (for example, average) the per-image tokens before concatenating all input tokens. This pooled image tokens model achieved a CIDEr score of 56.3 for 4-shots COCO captioning, which is substantially lower than the full model's CIDEr score of 61.7. This highlights the importance of keeping all the tokens coming out of the ViT encoder, despite the computational overhead.

Limited-range Encoding Attention. We explore per-example image-text attention, as proposed and applied in [10]. Under this approach, the image query tokens for each example can only attend

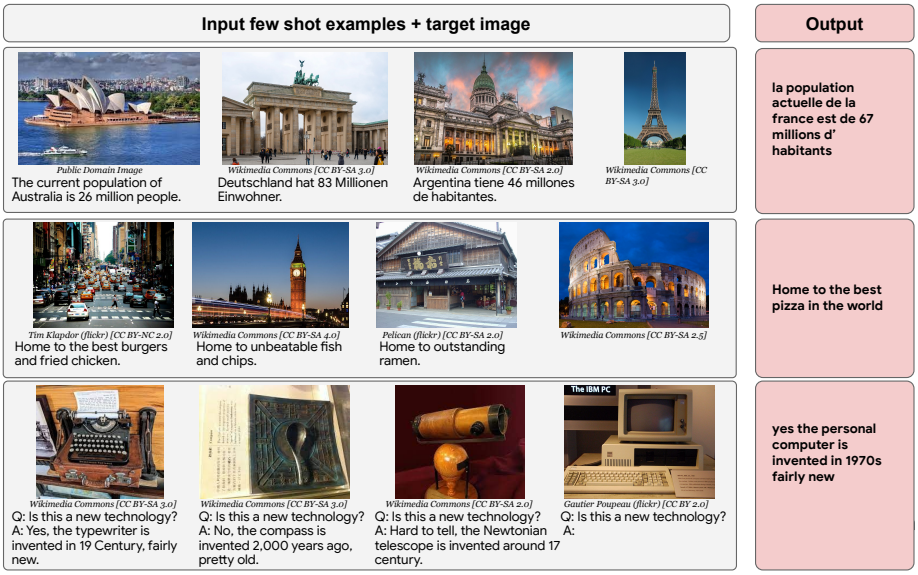


图 8：少样本图像描述（前两行）和 VQA（最后一行）任务的定性结果。

	Crossmodal-3600 Captioning							
	en	fr	hi	iw	ro	th	zh	35-lang avg.
PaLI-X 0-shot	48.8	25.0	10.5	20.1	13.0	33.3	18.4	22.7
PaLI-X (2 text-only shots[5])	54.5	46.7	12.0	22.2	9.4	40.3	23.7	25.8
PaLI-X 4 shots	77.8	62.5	22.2	38.7	30.2	56.0	27.7	45.1
PaLI-X 32 shots	81.4	66.1	25.6	40.6	32.4	59.4	29.7	47.1
PaLI-X (finetuned)	94.2	78.7	32.0	46.9	36.9	75.3	36.1	53.1

表 19：PaLI-X 模型在多语言图像描述任务上的少样本性能。

展示了少样本能力的跨多语言迁移。第二行用一个国家的流行美食为图像添加说明，表明少样本方法可以访问模型的世界知识。最后一行展示了一个带有类似解释场景的 VQA，我们询问图像中的技术是否“新”。一般来说，图中所示的个人电脑生产于 40 多年前，考虑到当前高科技发展的快速步伐，可以被视为旧技术。然而，3 个输入示例为“新”这一概念提供了详细校准，少样本模型成功理解了上下文，并针对这台非常老旧的 PC 输出“新”，同时给出了合理的解释。

B.5.3 少样本消融结果

在本节中，我们展示并讨论了我们为少样本设置所探索的一些消融结果，以指导我们对 PaLI-X 的最终设计选择。除非另有说明，我们使用一个具有相同编码器-解码器架构的 7 亿参数模型，该模型由 ViT-B/16 视觉编码器和 mT5-base 编码器-解码器语言模型组成。

池化与不池化图像 token 为了减轻多样本带来的计算负担，我们可以在拼接所有输入 token 之前，对每张图像的 token 进行池化（例如取平均）。这种池化图像 token 的模型在 4-shot COCO 图像描述任务中取得了 56.3 的 CIDEr 分数，显著低于完整模型 61.7 的 CIDEr 分数。这凸显了保留 ViT 编码器输出的所有 token 的重要性，尽管这会带来计算开销。

有限范围编码注意力。我们探索了按样本的图像-文本注意力，如 [10] 中所提出并应用的。在这种方法下，每个样本的图像查询标记只能关注

to its corresponding text tokens, while the text query tokens can attend to all tokens. By using this per-example attention model, we achieved a CIDEr score of 59.6, which is 2.1 points lower than the full attention model's CIDEr score of 61.7 for 4-shots COCO captioning.

Attention re-weighting for large number of shots. We report the few-shot results on COCO captioning from early-stopped PaLI-2 3B models; in this case, we did not apply normalized attention in training. We provide the test results with and without attention re-weighting during *inference* for a different number of encoder shots. Attention re-weighting achieves increasing CIDEr scores of 82.1, 84.3 and 84.5 with 4, 8 and 16 shots respectively. On the other hand, the model achieves 83.4, 76.5 and 66.3 without attention re-weighting. The decreasing performance may suggest that the model fails to locate the target image and text prompt among the large number of shots, whereas the attention re-weighting helps the model to focus on the target features. Accordingly, we decided to include attention re-weighting during finetuning for PaLI-X.

Distributing shots between encoder and decoder. We explore the use of both encoder and decoder shots during pre-training. We pretrain the PaLI-2 700M model on PaLI-2 mixtures with varying number of encoder shots (between 1 and 4). The remaining shots (up to exactly 4) are used as decoder shots. Using only encoder shots leads to a 64.0 CIDEr score for 4 shots in COCO captioning. The best mix of encoder and decoder shots achieves a CIDEr score of 65.2. This suggests splitting shots leads to a more challenging pre-train task that helps the model learn more efficiently.

B.6 Finetuning hyperparameters

The hyperparamter choices for downstream finetuning experiments are summarized in Table 20. As mentioned in the Main Text, for all of the downstream finetuning experiments, we used a reduced set of hyperparameters, without heavy per-task optimization.

Benchmark	learning rate schedule	Steps before LR decay to 0	batch size
COCO	linear decay from 1e-4	10k	256
VQAv2		20k	256
OCRVQA		20k	256
Multitask-VQA		20k	256
Multitask-Captioning		20k	256
All other		5k	128

Table 20: Hyperparameter used for finetuning PaLI-X.

B.7 Multi-task finetuning

We deduplicated every training set mixture over the test sets of every task in order to prevent leakage of any test-set examples into the training set. The mixture is formed by putting the training examples of each subtask together, with heuristic adjustments for a better balance. Following the resolutions for the single-task finetuning, the multi-task captioning and VQA finetuning are done with 672 and 756 image resolutions, respectively. The multitask finetuning covers just about 5M examples, which is 20k steps with a batch size of 256. For scene-text and document understanding tasks, the multi-task finetuning uses the end-to-end setting without OCR pipeline input.

The following aspects made multitask finetuning particularly challenging: (i) all tasks used the same prompt without task-specific indicators; the model is thus required to adapt to the style of multiple benchmarks simultaneously. 2) We do not perform per-task validation set optimization. All subtasks are evaluated using the same checkpoint, but tasks converge to their optimal value at a different pace.

B.8 Ablation studies

We first show in Table 22 the advantage brought by the OCR co-training stage of ViT-22B. We pair the vanilla ViT-22B and the ViT-22B with additional OCR co-training with a small language model mT5-base and pretrain these models on 40M of WebLI-OCR data with the splitOCR objective, before finetuning on ST-VQA. Co-training on image and OCR classification has a significant advantage on

对应的文本标记，而文本查询标记可以关注所有标记。通过使用这种逐样本注意力模型，我们在 4-shot COCO 图像描述任务中取得了 59.6 的 CIDEr 分数，比全注意力模型 61.7 的 CIDEr 分数低 2.1 分。

大量样本下的注意力重加权。我们报告了早停的 PaLI-2 3B 模型在 COCO 图像描述任务上的少样本结果；在这种情况下，我们没有在训练中应用归一化注意力。我们提供了在 *inference* 期间针对不同数量的编码器样本、使用和不使用注意力重加权的测试结果。使用注意力重加权时，4、8 和 16 个样本分别取得了 82.1、84.3 和 84.5 的 CIDEr 分数提升。另一方面，不使用注意力重加权时，模型分别取得了 83.4、76.5 和 66.3。性能下降可能表明模型无法在大量样本中定位目标图像和文本提示，而注意力重加权有助于模型聚焦于目标特征。因此，我们决定在 PaLI-X 的微调过程中加入注意力重加权。

在编码器和解码器之间分配示例。我们在预训练期间探索同时使用编码器示例和解码器示例。我们在 PaLI-2 混合数据上预训练 PaLI-2 700M 模型，并改变编码器示例的数量（介于 1 到 4 之间）。剩余的示例（总数恰好达到 4）用作解码器示例。在 COCO 图像描述任务中，仅使用编码器示例时，4 个示例的 CIDEr 得分为 64.0。编码器和解码器示例的最佳组合达到 65.2 的 CIDEr 得分。这表明拆分示例会带来更具挑战性的预训练任务，有助于模型更高效地学习。

B.6 微调超参数

下游微调实验的超参数选择总结见表 20。如正文所述，对于所有下游微调实验，我们使用了一组精简的超参数，而没有进行繁重的逐任务优化。

Benchmark	learning rate schedule	Steps before LR decay to 0	batch size
COCO	linear decay from 1e-4	10k	256
VQAv2		20k	256
OCRVQA		20k	256
Multitask-VQA		20k	256
Multitask-Captioning		20k	256
All other		5k	128

表 20：用于微调 PaLI-X 的超参数。

B.7 多任务微调

我们对每个任务的测试集上的所有训练集混合数据进行了去重，以防止任何测试集样本泄漏到训练集中。该混合数据通过将每个子任务的训练样本合并在一起形成，并通过启发式调整实现更好的平衡。沿用单任务微调的分辨率设置，多任务图像描述和 VQA 微调分别使用 672 和 756 的图像分辨率。多任务微调总共涵盖约 500 万个样本，即在批量大小为 256 的情况下训练 2 万步。对于场景文本和文档理解任务，多任务微调采用端到端设置，不输入 OCR 流水线结果。

以下几个方面使多任务微调尤其具有挑战性：(i) 所有任务都使用相同的提示词，没有任务特定的指示符；因此，模型需要同时适应多个基准的风格。2) 我们不进行按任务划分的验证集优化。所有子任务都使用同一个检查点进行评估，但各任务以不同的速度收敛到其最优值。

B.8 消融实验

我们首先在表 22 中展示 ViT-22B 的 OCR 联合训练阶段所带来的优势。我们将原始 ViT-22B 以及经过额外 OCR 联合训练的 ViT-22B 与小型语言模型 mT5-base 配对，并在 4000 万 WebLI-OCR 数据上使用 splitOCR 目标对这些模型进行预训练，然后在 ST-VQA 上进行微调。图像和 OCR 分类的联合训练在以下方面具有显著优势

Model	VQA v2	OK VQA	Text VQA	VizWiz VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	Chart QA	Avg.
Split	test-dev	val	val	test-dev	val	test	test	test	test	-
Previous Multi-task SOTA	84.3	64.5	68.4	71.6	75.1	71.3	40.0	76.6	70.5	-
Single-task FT	**86.0**	**66.1**	**71.9**	**72.6**	**80.2**	**75.9**	49.2	80.0	**70.9**	-
Multi-task FT	84.3	63.5	71.4	71.4	79.0	73.4	**50.7**	**80.9**	70.6	-
Multi-task (+/-)	-1.7	-2.6	-0.5	-1.2	-1.2	-2.4	+1.5	+0.9	-0.3	-0.8

Table 21: Scores from multi-task finetuning compared with those from single-task finetuning for VQA. Validation or test-dev set numbers are reported for some tasks.

ST-VQA and TextVQA. In the meantime, the performance on VQAv2, which is not very scene-text heavy, is improved as well. Moreover, we found that making the top left patch white, which helped the co-training of image classification and ocr classification on ViT-22B, is not required for the subsequent training of PaLI-X.

For ablation of the PaLI-X training procedure, we used a 5B model with UL2-3B and ViT-G with 2B parameters, which is roughly a 10:1 down-scale of the PaLI-X 55B model.

Model	OCR-task Indicator	ST-VQA	TextVQA	VQAv2	3-task avg.
mT5-base + Vanilla ViT-22B	No	42.6	36.1	68.9	49.2
mT5-base + ViT-22B-OCR	No	**47.0**	38.9	69.8	**51.9**
mT5-base + ViT-22B-OCR	Yes	46.2	**39.4**	**70.2**	**51.9**

Table 22: Advantage of the OCR co-training stage of ViT-22B. Pretraining is performed with resolution 224×224 and finetuning is with 448×448. Numbers reported are on validation split.

For stage 1 training, we show in Table 23 that adding image token generation does not harm the performance on the main image+language understanding tasks.

Mixture	COCO	VQAv2
without ViT-VQGAN	139.3	77.3
with 10% ViT-VQGAN	139.7	77.1

Table 23: Ablation experiment showing adding ViT-VQGAN tokens does not harm understanding performance (captioning and VQA tasks).

Model	VQA v2	OK VQA	Text VQA	VizWiz VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	Chart QA	Avg.
Split	test-dev	val	val	test-dev	val	test	test	test	test	-
Previous Multi-task SOTA	84.3	64.5	68.4	71.6	75.1	71.3	40.0	76.6	70.5	-
Single-task FT	**86.0**	**66.1**	**71.9**	**72.6**	**80.2**	**75.9**	49.2	80.0	**70.9**	-
Multi-task FT	84.3	63.5	71.4	71.4	79.0	73.4	**50.7**	**80.9**	70.6	-
Multi-task (+/-)	-1.7	-2.6	-0.5	-1.2	-1.2	-2.4	+1.5	+0.9	-0.3	-0.8

表21：VQA中多任务微调与单任务微调的得分对比。部分任务报告的是验证集或test-dev集上的数值。

ST-VQA 和 TextVQA。与此同时，在场景文本占比并不高的 VQAv2 上，性能也有所提升。此外，我们发现，将左上角图像块设为白色曾有助于在 ViT-22B 上进行图像分类和 OCR 分类的协同训练，但对于 PaLI-X 的后续训练并不是必需的。

对于 PaLI-X 训练流程的消融实验，我们使用了一个 5B 模型，其中包含 3B 参数的 UL2-3B 和 2B 参数的 ViT-G，这大约是 PaLI-X 55B 模型按 10:1 缩小后的规模。

Model	OCR-task Indicator	ST-VQA	TextVQA	VQAv2	3-task avg.
mT5-base + Vanilla ViT-22B	No	42.6	36.1	68.9	49.2
mT5-base + ViT-22B-OCR	No	**47.0**	38.9	69.8	**51.9**
mT5-base + ViT-22B-OCR	Yes	46.2	**39.4**	**70.2**	**51.9**

表 22：ViT-22B 的 OCR 协同训练阶段的优势。预训练使用 224×224 分辨率进行，微调使用 448×448。报告的数值基于验证集。

对于第一阶段训练，我们在表23中表明，加入图像标记生成不会损害主要图像+语言理解任务上的性能。

Mixture	COCO	VQAv2
without ViT-VQGAN	139.3	77.3
with 10% ViT-VQGAN	139.7	77.1

表 23：消融实验表明，添加 ViT-VQGAN 标记不会损害理解性能（图像描述和 VQA 任务）。

C Additional results: Video Captioning and QA

Below we give a brief description of each video data set we used for evaluation. Note that we freshly collected the data when performing the experiments, which led to different effective numbers of videos in different splits in some cases, see Table 24.

These descriptions refer to the original dataset size, but we train on (sometimes significantly) fewer videos — the exact numbers are given in Table 24. This is because not all videos in the datasets were available online at the time of writing (e.g., due to user deletion).

C.1 Datasets & Benchmarks

MSR-VTT [55]: This dataset consists of 10K open domain video clips for video captioning, with 20 captions each. The duration of each video clip is between 10 and 30 seconds. We follow the standard splits proposed by [55] and report results on the test set.

VATEX [56]: VATEX includes captions for 41K videos sampled from the Kinetics-600 dataset, with 10 English captions each. We report results on the English public test set.

ActivityNet Captions [57]: This dataset consists of 100K temporally localized sentences for 20k videos. We follow the standard split containing 50/25/25% of the dataset for training, validation and testing, and use ground truth temporal proposals at evaluation following [57]. Note that following other works [62], we use the `val_1` split for validation and `val_2` split for testing.

Spoken Moments in Time (SMIT) [58]: This dataset consists of long captions obtained via audio recordings for 500k short video clips. While this dataset has been traditionally only used for text to video retrieval, we find that it is a strong benchmark for captioning as it is the largest manually annotated set of videos with text captions.

ActivityNet-QA [61]: The dataset contains 58,000 question-answer pairs for videos in the ActivityNet dataset [97]. We report accuracy (using exact string match) on the test split. Note that we do open-ended generation for all VideoQA datasets.

MSR-VTT-QA [60]: This dataset was created using a semi-automatic pipeline on top of the MSR-VTT dataset. We report accuracy (using exact string match) on the test split.

NExT-QA [59]: We focus on the Open-Ended QA task, which consists of 52,044 question-answer pairs for a total of 5,440 videos (sampled from the VidOr dataset[98]). Exactly following Next-QA [59] and Flamingo [10], we report the Wu-Palmer Similarity (WUPS) on the test set.

		MSR-VTT	VATEX	ANet-Cap	SMIT	M-V-QA	ANet-QA	NExT-QA
Original size	train	6513	25991	37421	481094	158581	32000	37523
	valid.	497	3000	17505	14604	12278	18000	5343
	test	2990	6000	17031	3513	72821	8000	9178
Dataset size	train	4768	22902	30982	481094	116943	28020	37523
	valid.	327	2657	14604	8096	8215	15890	5343
	test	2144	5276	14234	3513	53014	7050	9178
% Remaining	train	73.21	88.12	82.79	100.00	73.74	87.56	100.00
	valid.	65.79	88.57	83.43	100.00	66.91	88.28	100.00
	test	71.71	87.93	83.58	100.00	72.80	88.13	100.00

Table 24: We freshly collect the data sets from the respective data sources. In cases where there are multiple question-answer pairs per video we report the number of question-answer pairs. Similarly, for ActivityNet Captions we report the number of captions. Due to missing videos which were removed after the original data sets were defined, most of our data sets are missing 10% of the videos or more.

C 附加结果：视频字幕生成和问答

下面我们简要描述用于评估的每个视频数据集。请注意，我们在进行实验时重新收集了数据，这导致在某些情况下不同划分中的有效视频数量有所不同，见表 24。

这些描述指的是原始数据集大小，但我们训练所用的视频数量（有时显著）更少——确切数量见表 24。这是因为在撰写本文时，并非数据集中的所有视频都可在线获取（例如，由于用户删除）。

C.1 数据集与基准测试

MSR-VTT [55]：该数据集包含 1 万个用于视频字幕生成的开放域视频片段，每个片段配有 20 条字幕。每个视频片段的时长在 10 到 30 秒之间。我们遵循 [55] 提出的标准划分，并在测试集上报告结果。

VATEX [56]：VATEX 包含从 Kinetics-600 数据集中采样的 4.1 万个视频的描述，每个视频有 10 条英文描述。我们在英文公开测试集上报告结果。

ActivityNet Captions [57]：该数据集包含针对 2 万个视频的 10 万个时间定位句子。我们遵循标准划分，将数据集的 50/25/25% 分别用于训练、验证和测试，并在评估时按照 [57] 使用真实时间提案。请注意，按照其他工作 [62]，我们使用 val_1 划分进行验证，使用 val_2 划分进行测试。

时间中的口语瞬间（SMIT）[58]：该数据集由通过音频录制为 50 万个短视频片段获取的长字幕组成。虽然该数据集传统上仅用于文本到视频检索，但我们发现它也是一个强大的字幕生成基准，因为它是规模最大的人工标注视频文本字幕数据集。

ActivityNet-QA [61]：该数据集包含 ActivityNet 数据集 [97] 中视频的 58,000 个问答对。我们报告测试集上的准确率（使用精确字符串匹配）。请注意，我们对所有 VideoQA 数据集都采用开放式生成。

MSR-VTT-QA [60]：该数据集是在 MSR-VTT 数据集基础上使用半自动流程创建的。我们在测试集划分上报告准确率（使用精确字符串匹配）。

NExT-QA [59]：我们关注开放式问答任务，该任务包含 52,044 个问答对，覆盖总计 5,440 个视频（从 VidOr 数据集 [98] 中采样）。严格遵循 NExT-QA [59] 和 Flamingo [10]，我们在测试集上报告 Wu-Palmer 相似度（WUPS）。

		MSR-VTT	VATEX	ANet-Cap	SMIT	M-V-QA	ANet-QA	NExT-QA
Original size	train	6513	25991	37421	481094	158581	32000	37523
	valid.	497	3000	17505	14604	12278	18000	5343
	test	2990	6000	17031	3513	72821	8000	9178
Dataset size	train	4768	22902	30982	481094	116943	28020	37523
	valid.	327	2657	14604	8096	8215	15890	5343
	test	2144	5276	14234	3513	53014	7050	9178
% Remaining	train	73.21	88.12	82.79	100.00	73.74	87.56	100.00
	valid.	65.79	88.57	83.43	100.00	66.91	88.28	100.00
	test	71.71	87.93	83.58	100.00	72.80	88.13	100.00

表24：我们从相应的数据源重新收集了这些数据集。对于每个视频包含多个问答对的情况，我们报告问答对的数量。类似地，对于 ActivityNet Captions，我们报告字幕的数量。由于在原始数据集定义之后有些视频被移除而缺失，我们的大多数数据集缺少了10%或更多的视频。

D Additional results: Image Classification

Setup for zero-shot and finetuning evaluation The setup used for the experiments here uses the PaLI-X model to generate directly the (English) class name using the captioning prompt. The output is considered correct if it matches exactly the class name (apart from ImageNet-REAL, where we check if the class corresponding to the output is in the set of correct labels).

Zero-shot Evaluation results We use the same scoring technique as in PaLI [5] to evaluate PaLI-X in zero-shot setting (without training on any Imagenet data). We use the PaLI-X model obtained after the first stage of training (using the base 224 image resolution).

The results are presented in Table 25. We compare the results to PaLI [5] - previous zero-shot generative SOTA, and Flamingo [10] - another generative model of similar architecture with comparable 1-shot and 5-shot results. Overall, we report that the results between PaLI and PaLI-X for 0-shot are similar.

Model (ImageNet data)	INet	REAL	INet-R	INet-A	INet-Sketch	INet-v2	ObjNet
Flamingo-80B (1-shot)	71.9	-	-	-	-	-	-
Flamingo-80B (5-shot)	77.3	-	-	-	-	-	-
PaLI (17B) (0-shot)	**72.11**	**76.43**	81.97	44.70	**63.83**	**64.46**	42.62
PaLI-X (0-shot)	71.16	75.75	**82.96**	**46.13**	61.58	63.91	**44.58**

Table 25: Top 1 accuracy results of 0-shot image classification on ImageNet [66], ImageNet-REAL [67], ImageNet-R [68], ImageNet-A [69], ImageNet-Sketch [70], Imagenet-v2 [71] and ObjectNet [99].

Finetuning To test image classification capabilities, we finetune PaLI-X on ImageNet [66] and evaluate the resulting model on ImageNet-REAL [67] and out-of-distribution datasets: ImageNet-R [68], ImageNet-A [69], ImageNet-Sketch [70], ImageNet-v2 [71].

We use the model from the first training stage (at resolution 224) and the one from the last training stage (at resolution 756). We use the same training hyperparameters for all of runs (selected without any hyperparameter tuning).

The results can be seen in Table 26. We compare the results to generative model with open vocab – GiT2 [9] (using 384 image resolution), which is the current SOTA for full-finetuning on ImageNet. PaLI-X achieves close to SOTA results for generative models on Imagenet, and other datasets.

Model (resolution)	INet	REAL	INet-R	INet-A	INet-Sketch	INet-v2
GIT2 (384)	**89.22**	-	-	-	-	-
PaLI 3B (224)	85.11	88.71	**81.11**	45.71	70.00	78.23
PaLI 17B (224)	86.13	88.84	78.21	50.00	71.21	78.91
PaLI-X (224)	88.22	90.36	77.66	55.97	72.56	81.42
PaLI-X (756)	88.82	90.80	79.97	**73.47**	**73.39**	83.48
PaLI-X † (756)	89.19	**90.98**	80.06	72.57	73.37	**83.66**

Table 26: Classification (top-1) accuracy with Imagenet [66] fine-tuning on: ImageNet, ImageNet-REAL [67], ImageNet-R [68], ImageNet-A [69], ImageNet-Sketch [70], Imagenet-v2 [71] (resolution in parentheses). PaLI-X † fine-tuned for 2.2x more steps.

D 附加结果：图像分类

零样本和微调评估设置 此处实验所使用的设置采用 PaLI-X 模型通过字幕生成提示直接生成（英文）类别名称。如果输出与类别名称完全匹配，则认为输出正确（ImageNet-REAL 除外，在该数据集中我们会检查输出对应的类别是否在正确标签集合中）。

零样本评估结果 我们使用与 PaLI [5] 中相同的评分技术，在零样本设置下评估 PaLI-X（未在任何 Imagenet 数据上进行训练）。我们使用第一阶段训练后获得的 PaLI-X 模型（使用基础的 224 图像分辨率）。

结果见表 25。我们将结果与 PaLI [5]（此前的零样本生成式 SOTA）以及 Flamingo [10]（另一种架构相似的生成式模型，其 1-shot 和 5-shot 结果具有可比性）进行了比较。总体而言，我们报告 PaLI 和 PaLI-X 在 0-shot 设置下的结果相近。

Model (ImageNet data)	INet	REAL	INet-R	INet-A	INet-Sketch	INet-v2	ObjNet
Flamingo-80B (1-shot)	71.9	-	-	-	-	-	-
Flamingo-80B (5-shot)	77.3	-	-	-	-	-	-
PaLI (17B) (0-shot)	**72.11**	**76.43**	81.97	44.70	**63.83**	**64.46**	42.62
PaLI-X (0-shot)	71.16	75.75	**82.96**	**46.13**	61.58	63.91	**44.58**

表 25：在 ImageNet [66]、ImageNet-REAL [67]、ImageNet-R [68]、ImageNet-A [69]、ImageNet-Sketch [70]、Imagenet-v2 [71] 和 ObjectNet [99] 上的 0-shot 图像分类 Top 1 准确率结果。

微调 为了测试图像分类能力，我们在 ImageNet [66] 上对 PaLI-X 进行微调，并在 ImageNet-REAL [67] 以及分布外数据集 ImageNet-R [68]、ImageNet-A [69]、ImageNet-Sketch [70]、ImageNet-v2 [71] 上评估所得模型。

我们使用第一训练阶段的模型（分辨率为 224）以及最后训练阶段的模型（分辨率为 756）。我们在所有运行中使用相同的训练超参数（未进行任何超参数调优而选定）。

结果见表 26。我们将结果与开放词汇生成模型 GiT2 [9]（使用 384 图像分辨率）进行比较，后者是当前在 ImageNet 上进行全量微调的 SOTA。PaLI-X 在 ImageNet 以及其他数据集上取得了接近生成模型 SOTA 的结果。

Model (resolution)	INet	REAL	INet-R	INet-A	INet-Sketch	INet-v2
GIT2 (384)	**89.22**	-	-	-	-	-
PaLI 3B (224)	85.11	88.71	**81.11**	45.71	70.00	78.23
PaLI 17B (224)	86.13	88.84	78.21	50.00	71.21	78.91
PaLI-X (224)	88.22	90.36	77.66	55.97	72.56	81.42
PaLI-X (756)	88.82	90.80	79.97	**73.47**	**73.39**	83.48
PaLI-X † (756)	89.19	**90.98**	80.06	72.57	73.37	**83.66**

表26：在以下数据集上使用 ImageNet [66] 微调的分类（top-1）准确率：ImageNet、ImageNet-REAL [67]、ImageNet-R [68]、ImageNet-A [69]、ImageNet-Sketch [70]、ImageNet-v2 [71]（括号内为分辨率）。PaLI-X † 微调步数增加了 2.2 倍。

E Object Detection

E.1 Object detection as a VLM task

Object detection is framed similarly to Pix2seq [72], with two key differences: the use of a natural language vocabulary, and class-conditioning. Prompt classes are fed to PaLI-X's text encoder, in the format `detect class1 and class2 and class3`. The model is trained to only output bounding boxes corresponding to classes in this prompt. We represent bounding boxes as coordinates in the same style as pix2seq [72]; that is, 4 integers y_{min} x_{min} y_{max} x_{max} ranging from 0 to 999. Figure 9 shows an example input.

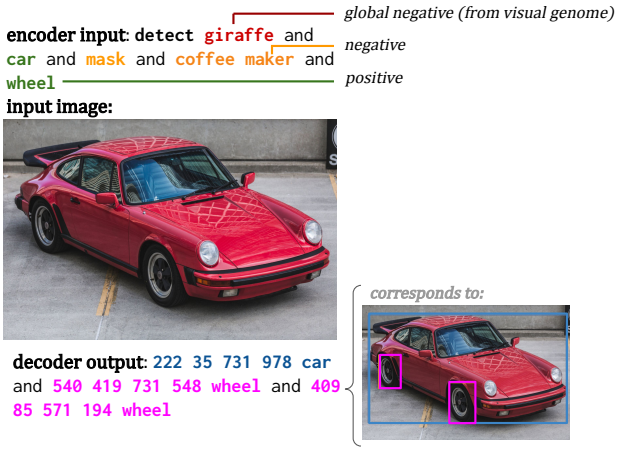


Image credits: Matthew Henry, burst, https://burst.shopify.com/photos/vintage-red-porsche

Figure 9: An example training pair, consisting of the text prompt, the image and the expected output. The prompt consists of multiple classes; we show a hypothetical Open Images V4 example, with positives 'car' and 'wheel', negative 'giraffe' and global negatives 'mask' and 'coffee maker' (sampled from the visual genome label space).

Prompt sampling hyperparameters During training, a prompt for each example. We construct prompts from three pieces of information:

- *Positives*: These are the bounding boxes for objects definitely present in the image. During training, per example we sample $p^+ \sim \mathcal{U}(0, P^+_{\max})$, and keep that proportion of positives.
- *Negatives*: These are the known instance negatives i.e. bounding boxes for objects definitely not present. For exhaustively labelled datasets like COCO, this is simply classes not labelled as positives. For non-exhaustively labelled datasets like LVIS, these are the classes not labelled as positives, which were presented to raters. During training sample $f^- \sim \mathcal{U}(0, 5.0)$, and use up to $f^- \times n^+$, where n^+ is the number of positives after sampling p^+.
- *Global negatives*: These are negatives which are not explicitly labelled as negatives. They are taken from a wider label space combining multiple detection datasets. For a given example, valid global negatives consist of classes from the wider label space not explicitly labelled as positives or negatives. During training, we sample $f^{GN} \sim \mathcal{U}(0, 5.0)$ and append $f \times n^+$ global negatives, where n_+ is the number of positives after sampling p^+.

  By default, the combined label spaces of Visual Genome, Objects365 and OpenImagesV4 was used as the global label space, with the exception of detection finetuning, where LVIS and COCO label spaces were also added.

We truncate the number of total classes to $n_{\max}$. $n_{\max}$ and $P^+_{\max}$) are tuned per dataset to meet sequence lengths. Afer truncatation, we shuffle classes in the prompt.

E 目标检测

E.1 将目标检测作为 VLM 任务

目标检测的表述方式与 Pix2seq [72] 类似，但有两个关键区别：使用自然语言词汇表，以及类别条件化。提示类别会以 detect class1 and class2 and class3 的格式输入到 PaLI-X 的文本编码器中。模型被训练为只输出与该提示中的类别对应的边界框。我们以与 pix2seq [72] 相同的风格表示边界框坐标；也就是说，使用 4 个整数 y_{min} x_{min} y_{max} x_{max}，取值范围为 0 到 999。图 9 展示了一个输入示例。

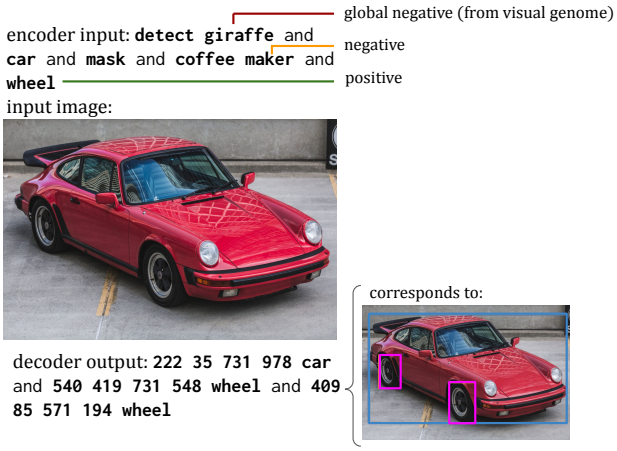


Image credits: Matthew Henry, burst, https://burst.shopify.com/photos/vintage-red-porsche

图 9：一个训练样本对示例，包括文本提示、图像和预期输出。该提示包含多个类别；我们展示了一个假设的 Open Images V4 示例，其中正类为“汽车”和“车轮”，负类为“长颈鹿”，全局负类为“面具”和“咖啡机”（从 Visual Genome 标签空间中采样）。

提示采样超参数 在训练期间，会为每个示例采样一个提示。我们根据三部分信息构建提示：

- *Positives*：这些是图像中确定存在的对象的边界框。在训练期间，对于每个样本，我们采样 $p^+ \sim \mathcal{U}(0, P^+_{\max})$，并保留该比例的正样本。
- *Negatives*：这些是已知的实例负类，即明确不存在对象的边界框。对于像 COCO 这样穷尽标注的数据集，这就是未标注为正类的类别。对于像 LVIS 这样非穷尽标注的数据集，这些是未标注为正类、但已呈现给标注者的类别。在训练期间采样 $f^- \sim \mathcal{U}(0, 5.0)$，并最多使用 $f^- \times n^+$，其中 n^+ 是采样 p^+ 后的正类数量。
- *Global negatives*：这些是未被明确标注为负类的负样本。它们来自结合了多个检测数据集的更广泛标签空间。对于给定示例，有效的全局负类由更广泛标签空间中未被明确标注为正类或负类的类别组成。在训练期间，我们采样 $f^{GN} \sim \mathcal{U}(0, 5.0)$ 并追加 $f \times n^+$ 个全局负类，其中 n_+ 是在采样 p^+ 后的正类数量。默认情况下，将 Visual Genome、Objects365 和 OpenImagesV4 的组合标签空间用作全局标签空间，但检测微调除外，在该情况下还会加入 LVIS 和 COCO 标签空间。

我们将总类别数截断为 $n_{\max}$。$n_{\max}$ 和 $P^+_{\max}$) 会针对每个数据集进行调优，以满足序列长度要求。截断后，我们会在提示中打乱类别顺序。

E.2 Preprocessing

During pre-training, data is preprocessed to remove all LVIS-rare labels, following the protocol of OwlViT [28]. This is not done for detection finetuning. Images are randomly flipped horizontally, and randomly resized to between 0.3 and 2.0 $\times$ their original sized, followed by selecting a random square crop of the current training resolution. If the image is resized to be smaller than the current resolution, it is left as is. Images are finally padded to a square.

E.3 Licenses and attribution for images used in Main Text Figure 2

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E.2 预处理

在预训练期间，按照 OwlViT [28] 的协议，对数据进行预处理以移除所有 LVIS-rare 标签。检测微调时不进行此操作。图像会被随机水平翻转，并随机缩放到其原始尺寸的 0.3 到 2.0 × 之间，随后在当前训练分辨率下随机选择一个正方形裁剪区域。如果图像被缩放到小于当前分辨率，则保持不变。最后将图像填充为正方形。

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